NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

A STATE UNIVERSITY

UNDER DELHI ACT 06 OF 2018, GOVT. OF NCT OF DELHI

Azad Hind Fauj Marg, Sector-3, Dwarka, New Delhi-110078

SCHEME OF COURSES AND EXAMINATION

FOR

BACHELOR OF TECHNOLOGY

MECHANICAL ENGINEERING

(Effective from the Session: 2019-202)

APPROVED BY

The Senate in its II to VII meetings

The Board of Management

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1. INTRODUCTION

NSUT has embarked on its journey towards excellence in academics through the introduction of a novel system of learning that is being followed in many reputed universities globally. The Choice Based Credit System (CBCS) has been proposed by University Grants Commission (UGC) on recommendations of the National Knowledge Commission, to improve the quality of higher education in India. NSUT proposes to adopt CBCS for its Bachelor of Technology courses

CBCS is the mother of student centric educational reforms. A student is provided with an academically rich, highly flexible learning system blended with abundant provision for skill practice and activity orientation that he/she could learn in depth without sacrificing his/her creativity. A student can exercise the option to decide his/her own pace of learning- slow, normal or accelerated plan and sequence his/her choice of paper, learn to face challenges through term work/ project work and may venture out to acquire extra knowledge/ proficiency through add- on facilities. The great advantage of CBCS is that the learning process is made continuous and the evaluation process is not only made continuous but also made learner-centric and is designed to recognize the capability and talent of a student.

2. CURRICULUM STRUCTURE

B.Tech. programme of the University shall be based upon CBCS and shall have well defined Programme Educational Objectives (PEOs). All the courses shall have well-defined Course Outcomes (COs). Courses shall be of three kinds: Core, Elective and Foundation.

a. **Core Course (CC):** This is a course which is to be compulsorily studied by a student as a core requirement to complete the requirements of the B.Tech. programme.

b. **Elective Course:** This is a course which can be chosen from a pool of elective courses. It is intended to support the discipline of study by providing an expanded scope, enabling exposure to another discipline/domain and nurturing a student's proficiency and skill. An elective may be of the following types:

   i. **Discipline Centric Elective (ED):** It is an elective course that adds proficiency to

the students in the discipline.

ii. **Generic Elective (EG):** It is an elective course taken from other engineering subjects and enhances the generic proficiency and interdisciplinary perspective of students.

iii. **Open Elective (EO):** It is an elective course taken from a common pool of non-engineering disciplines that broadens the perspective of an engineering student. These electives shall comprise two groups: Open electives of the Humanities, Social Sciences and Management group and Open electives of the Sciences group.

c. **Foundation Course:** A Foundation course leads to knowledge enhancement and provides value-based training. Foundation courses may be of two kinds:

i. **Compulsory Foundation (FC):** It is based upon the content that leads to fundamental knowledge enhancement in Sciences, Humanities, Social Sciences and Basic engineering. They are mandatory for all disciplines.

ii. **Elective Foundation (FE):** It can be taken from among a common pool of foundation courses which aim at value-based education. They may provide hands-on training to improve competencies, skills or provide education on human, societal, environmental and national values. These shall be mandatory, non-credit courses, which do not carry any credits but a student has to pass in order to be eligible for award of degree.

2.1 EVALUATION AND ASSESSMENT

The performance of a student in a semester shall be evaluated through continuous class assessment, mid semester examination (MSE) and end semester examination (ESE). Both the MSE and ESE shall be University examinations and will be conducted as notified by the Controller of Examination (CoE) of the University. The marks for continuous assessment (Sessional marks) shall be awarded at the end of the semester. The continuous assessment shall be based on class tests, assignments/tutorials, quizzes/viva-voce and attendance etc. The MSE/ESE shall comprise of written papers, practicals and viva-voce, inspection of certified course work in classes and laboratories, project work, design reports or by means of any combination of these methods.

The weightage of each of these modes of evaluation for the different types of courses

shall be as per Table 1. Further, the mechanism for continuous assessment shall be as per Table 2.

Table-1: Evaluation Scheme

S. No.	Type of Course	Continuous Assessment (CA) Theory	Mid-Semester Examination (MSE) Theory	End-Semester Examination (ESE) Theory	Continuous Assessment (CA) Practical	End-Semester Examination (ES) Practical
1	FE courses	Continuous Assessment only (100 marks)				
2	CC/FC/ED/EG/EO Theory with Tutorial	25	25	50	Nil	Nil
3	CC/FC/ED/ EG/EO Theory with Practical	15	15	40	15	15
4	Project I and Project II	Nil	Nil	Nil	40	60
5	Training	Nil	Nil	Nil	40	60
6	Work shop based Course	30	--	20	30	20
7	Audit Courses*	-	-	-	-	-
*The distribution of marks of practical and/or theory components for Audit courses shall be determined by the respective Departments.						

Table 2: Continuous Assessment

S. No.	Type of Course	Continuous Assessment (CA)
1	CC/FC/ED/EG/EO Theory with Tutorial	Two class tests, Assignments, Teachers' assessment (quizzes, viva-voce, attendance)
	CC/FC/ED/EG/EO Theory with Practical	One class test, One Lab test, Assignments/Projects, Teachers' assessment
2	FE courses ***	Two class tests, Assignments, Teachers' assessment
3	Project I /II	Mid-Semester Presentation, Report, Supervisor's Assessment
4	Training	As specified by the Department
5	Audit Courses	As specified by the Department

*** Foundation Elective Courses are value-based courses which may enhance the proficiency /skill. These electives could be communication skills, Spoken English, soft skills, Business and Management courses, entrepreneurship development, Knowledge of an

additional Foreign Language, Personality Development through sports, music, theatre, dance, etc.

The University provides to the students a pool of Foundation elective courses which may be offered by the following departments of the University:

i) Department of Humanities
ii)Department of Management
iii)Department of Personality Development

Note for Undergraduate students--

i. An Undergraduate student will have the liberty to choose any three-foundation elective course to study from the given list.
ii. He/ She can take only one foundation elective course in an ongoing semester.
iii. The study and clearing of foundation elective course is to be done by the end of 2nd year (fourth semester).
iv. For getting a Degree, it is mandatory to clear the entire three chosen foundation elective course.
v. Foundation elective courses are auditable course and there is no credits awarded to the students.

Note for Course Teacher--

The evaluation of the student is done through continuous assessment.

[Subject having Theory only] -- The course teacher evaluate through TWO class tests (25 marks each), ONE Assignment/Project (40 marks) and internal evaluation [one such component is attendance] (10 marks).

[Subject having Theory and Practical] -- Here, a course teacher evaluate for theory part through TWO class test (20 marks each) and internal evaluation [one such component is attendance] (10 marks). Similarly, for practical part ONE practical test (40 marks) and internal evaluation [one such component is attendance] (10 marks)

[Subject having Practical only] -- The course teacher takes TWO practical test (45 marks each) and internal evaluation [one such component is attendance] (10 marks)

2.2 SEMESTER WISE COURSE/CREDIT DISTRIBUTION

Table 3: Distribution of courses and credits

Semester	Types of courses as per NSUT Nomenclature						Courses/credits	Credits
	FE (NON-CREDIT)	FC	CC	ED	ED/EG/EO/ EO- Sciences / EO-SS & Mgmt	Training Project etc.		
I	01	05	00	00	00	00	06 courses 20 credits	**84 credits**
II	00	03	03	00	00	00	06 courses 24 credits	
III	01	00	05	00	00	00	06 courses 20 credits	
IV	01	00	05	00	00	00	06 courses 20 credits	
V	00	00	04	--	--	00	04 –07 courses 16-28 credits	**86 credits**
VI	00	00	04	--	-	00	04 –07 courses 16-28 credits	
VII	00	00	00	--	--	06	00 – 05 courses 06-26 credits	
VIII	00	00	00	--	--	08	00 – 05 courses 08-28 credits	
TOTAL CREDITS								**170**

- **ED :At least 4 courses (16 credits)**
- **EO- Sciences :At least 1 courses (04 credits)**
- **EO-SS & Mgmt :At least 2 courses (08 credits)**

2.3 COURSE CODE NOMENCLATURE

The courses of various B.Tech. programmes shall be assigned a course code as per the following nomenclature

2.3.1 COURSE/DEPARTMENT/SPECIALIZATION/BRANCH CODING

The courses of various B.Tech programmes shall be assigned a course code as per the defined nomenclature (Given later). This nomenclature shall use course/department/specialization/branch coding which are defined as given below.

TABLE 4: COURSE/DEPARTMENT CODES

XX	**Course Category Code**	FC	Foundation Core
		FE	Foundation Elective
		EO	Open Elective
YY	**Name of Department Code**	BT	Bio Technology
		CH	Chemistry
		CP	Computer Engineering, East Campus
		CS	Computer Science & Engineering
		CW	Civil Engg, West Campus
		EE	Electrical Engineering
		EC	Electronics & Communication Engineering
		EP	Electronics & Communication Engineering, East Campus
		HS	Humanities
		IC	Instrumentation & Control Engineering
		IT	Information Technology
		IW	Information Technology, West Campus
		ME	Mechanical Engineering
		MG	Management
		MP	Manufacturing Processes & Automation
		MT	Mathematics
		MW	Mechanical Engineering, West Campus
		PD*	Personality Development
		PH	Physics
Note: Second Alphabet P indicates East Campus, and W indicates West Campus			

TABLE 5: B.TECH SPECIALIZATION/BRANCH CODES

ZZ	BT	Bio Technology
	CA	Computer Science & Engineering with Artificial Intelligence
	CB	Computer Science and Engineering (Big Data Analytics) (CSDA)(NSUT EAST CAMPUS)
	CD	Computer Science and Engi)neering (Data Science) (CSDS)
	CE	Civil Engineering (CE)(NSUT WEST CAMPUS)
	CG	Geoinformatics (GI)(NSUT WEST CAMPUS)
	CI	Computer Science and Engineering (IOT) (CIOT)(EAST CAMPUS)
	CM	Maths & Computing
	CO	Computer Science & Engineering, Main Campus
	EA	Electronics and Communication Engineering (Artificial Intelligence and Machine Learning) (ECAM) (NSUT EAST CAMPUS)
	EC	Electronics & Communication Engineering
	EI	Electronics & Communication Engineering (Internet of Things)
	EE	Electrical Engineering
	IC	Instrumentation & Control Engineering
	II	Information Technology (Internet of Things) (IIOT)(NSUT WEST CAMPUS)
	IN	Information Technology (NetwoRk security)
	IT	Information Technology
	ME	Mechanical Engineering
	MP	Manufacturing Processes & Automation
	MV	Mechanical Engineering (Electric Vehicles) (MEEV)(NSUT WEST CAMPUS)

2.3.2 B.TECH COURSE CODE NOMENCLATURE

FOUNDATION CORE AND ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department Code			Course No.	
X	X	Y	Y	0	*	*

** can take numeric values only

XX and YY maybe chosen as given in Tables 1,2:

*PD offers FE courses like music, dance, yoga, sports, NSS, etc. A BOS for FE courses of PD nature (like Music, Dance, Yoga, NSS, etc), has be constituted with Dean Academics as the chairperson.

OTHER CORE AND ELECTIVE COURSES:

Program Code		Offering Department Code		Course Category	Course No.	
Z	Z	Y	Y	C/E	*	*

** can take numeric only;

C for Core and E for Elective (Discipline Centric);
YY and ZZ maybe chosen as given in Tables 1,2.

2.3.3 MOOCS (NPTEL BASED) FOUNDATION ELECTIVE COURSES AND OPEN ELECTIVE COURSES:

Course Category		Offering Department (NPTEL) Code		UG/PG	Course No.	
X	X	F	F	G	*	*

** can take numeric only;

XX	**Course Category Code**	FE	Foundation Elective
		EO	Open Elective
FF	**Name of Offering Department Code for NPTEL**	NH	Humanities & Social Sciences
		NM	Management
		NP	Personality Development
		NS	Sciences
G	**UG/PG**	0	B.Tech
		I	M.Tech

2.3.4 STUDENT ROLL NUMBER NOMENCLATURE:

Students shall be assigned roll numbers as given below.

1. B.Tech:

Year of Admission	U	ZZ (FROM TABLE 2)	4 DIGIT NUMBER

3 SEMESTER WISE COURSE ALLOCATION

3.1 COURSE ALLOCATION FOR SEMESTER I

Refer to the separate manual on Equivalent courses and scheme of courses with pre-requites if you want to find the linked courses and prerequisites for opting for EG courses.

B. TECH. (MECHANICAL ENGINEERING) SEMESTER I											
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
FCMT001	FC	Mathematics-I	3	1	0	4	25	25	50	-	-
FCHS005	FC	English	3	0	2	4	15	15	40	15	15
FCEC003	FC	Electrical and Electronics Engineering	3	0	2	4	15	15	40	15	15
FCCH008	FC	Environment Science & Green Chemistry	3	0	2	4	15	15	40	15	15
FCME006	FC	Basics of Mechanical Engineering	4	0	0	4	25	25	50	-	-
FEYY0--	FE	Foundation Elective	-	-	-	-	-	-	-	-	-
			23*			20					

1: One week induction program as per AICTE norms. Classes for I semester will commence one week later.
2*: The course codes, LTP distribution and Evaluation Scheme for Foundation Electives are given in Table 6. The actual weekly load depends upon the Foundation

Students of the Departments of Group I shall be offered courses as follows:

1. **Semester I: Computer Programming, Physics**
2. **Semester II: English, Environment Science and Green Chemistry**

Students of the Departments of Group II shall be offered courses as follows:

1. **Semester I: English, Environment Science and Green Chemistry**
2. **Semester II: Computer Programming, Physics**

3.2 COURSE ALLOCATION FOR SEMESTER II

B. TECH. (MECHANICAL ENGINEERING) SEMESTER II												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
FCCE002	FC	Computer Programming	3	0	2	4	15	15	40	15	15
FCMT007	FC	Mathematics-I	3	1	0	4	25	25	50	-	-
FCPH004	FC	Physics	3	0	2	4	15	15	40	15	15
MEMEC01	CC	Engineering Mechanics	3	0	2	4	15	15	40	15	15
MEMEC02	CC	Engineering Materials & Metallurgy	3	0	2	4	15	15	40	15	15
MEMEC03	CC	Engineering & Machine Drawing	2	0	4	4	-	-	-	40	60
			30			24					

3.3 COURSE ALLOCATION FOR SEMESTER III

B. TECH. (MECHANICAL ENGINEERING) SEMESTER III												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMTC04	CC	Numerical Methods & Computation	3	1	0	4	25	25	50	-	-
MEMEC05	CC	Manufacturing Processes-I	3	0	2	4	15	15	40	15	15
MEMEC06	CC	Strength of Materials	3	1	0	4	25	25	50	-	-
MEMEC07	CC	Thermal Engineering-I	3	0	2	4	15	15	40	15	15
MEMEC08	CC	Fluid Mechanics & Machines	3	0	2	4	15	15	40	15	15
FEYY0--	FE	Foundation Elective	-	-	-	-	-	-	-	-	-
			23*			20					
1*: The course codes, LTP distribution and Evaluation Scheme for Foundation Electives are given in Table 6. The actual weekly load depends upon the Foundation											

3.4 COURSE ALLOCATION FOR SEMESTER IV

B. TECH. (MECHANICAL ENGINEERING) SEMESTER IV												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMEC09	CC	Theory of Machines	3	0	2	4	15	15	40	15	15
MEMEC10	CC	Manufacturing Processes-II	3	0	2	4	15	15	40	15	15
MEMEC11	CC	Thermal Engineering-II	3	0	2	4	15	15	40	15	15
MEMEC12	CC	Industrial Engineering & Management	3	1	0	4	25	25	50	-	-
MEICC13	CC	Control Systems	3	0	2	4	15	15	40	15	15
FEYY0--	FE	Foundation Elective	-	-	-	-	-	-	-	-	-
			24*			20					

1*: The course codes, LTP distribution and Evaluation Scheme for Foundation Electives are given in Table 6. The actual weekly load depends upon the Foundation

3.5 COURSE ALLOCATION FOR SEMESTER V

B. TECH. (MECHANICAL ENGINEERING) SEMESTER V											
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMEC14	CC	Automobile Engineering	3	0	2	4	15	15	40	15	15
MEMEC15	CC	Robotics	3	0	2	4	15	15	40	15	15
MEMEC16	CC	Mechanical Design	3	1	0	4	25	25	50	-	-
MEEEC17	CC	Fundamentals & Operations of Electric Vehicles	3	1	0	4	25	25	50	-	-
MEMEE-- EOYY0--	ED/ EG/ EO	Elective	-	-	-	-	-	-	-	-	-
			18*			16-28					

1*: The course codes, LTP distribution and Evaluation Scheme for Program Electives are given in Table 7 (list under preparation). The actual weekly load depends upon the Elective.
2*: The course codes, LTP distribution and Evaluation Scheme for Open Electives are given in Table 8 (list under preparation). The actual weekly load depends upon the Elective.

3.6 COURSE ALLOCATION FOR SEMESTER VI

B. TECH. (MECHANICAL ENGINEERING) SEMESTER VI												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMEC18	CC	Artificial Intelligence in Mechanical Engineering	3	0	2	4	15	15	40	15	15
MEMEC19	CC	Computer Aided Design & Manufacturing (CAD & CAM)	3	0	2	4	15	15	40	15	15
MEICC20	CC	Mechatronics	3	0	2	4	15	15	40	15	15
MEMEE-- EOYY0--	ED/ EG/ EO	Elective	-	-	-	-	-	-	-	-	-
			15*			12-28					

1*: The course codes, LTP distribution and Evaluation Scheme for Program Electives are given in Table 7 (list under preparation). The actual weekly load depends upon the Elective.
2*: The course codes, LTP distribution and Evaluation Scheme for Open Electives are given in Table 8 (list under preparation). The actual weekly load depends upon the Elective.

3.7 COURSE ALLOCATION FOR SEMESTER VII

B. TECH. (MECHANICAL ENGINEERING) SEMESTER VII												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMEC21	CC	Training	0	0	0	2	-	-	-	30	70
MEMEC22	CC	Project - I	0	0	0	4	-	-	-	30	70
MEMEE-- EOYY0--	ED/ EG/ EO	Elective	-	-	-	-	-	-	-	-	-
			00*			06-28					
1*: The course codes, LTP distribution and Evaluation Scheme for Program Electives are given in Table 7 (list under preparation). The actual weekly load depends upon the Elective. 2*: The course codes, LTP distribution and Evaluation Scheme for Open Electives are given in Table 8 (list under preparation). The actual weekly load depends upon the Elective.											

3.8 COURSE ALLOCATION FOR SEMESTER VIII

B. TECH. (MECHANICAL ENGINEERING) SEMESTER VII												
Course Code	**Type**	**Courses**	**L**	**T**	**P**	**Credits**	**Evaluation Scheme (Percentage weights)**				
							Theory			**Practical**	
							CA	**MS**	**ES**	**CA**	**MS**
MEMEC23	CC	Project - II	0	0	0	8	-	-	-	30	70
MEMEE--EOYY0--	ED/EG/EO	Elective's	-	-	-	-	-	-	-	-	-
			00*			08-28					

1*: The course codes, LTP distribution and Evaluation Scheme for Program Electives are given in Table 7 (list under preparation). The actual weekly load depends upon the Elective.
2*: The course codes, LTP distribution and Evaluation Scheme for Open Electives are given in Table 8 (list under preparation). The actual weekly load depends upon the Elective.

4. SYLLABUS OF COURSES

4.1 SYLLABUS OF FOUNDATION COMPULSORY COURSES

Course No.	Title of the Course	Course Structure	Pre-requisite
FCMT001	**Mathematics - I**	**L-T-P: 3-1-0**	**None**

COURSE OUTCOMES (COs):

1. Analyze and test Infinite Series and its convergence,
2. Successive differentiation and expansion of the function,
3. Curvature and Radius of Curvature in different coordinate systems,
4. Applications of definite integral,
5. Consistency of system of equations, Eigenvalue and Eigenvector.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Infinite Series: Tests for convergence of series: p-series (with proof), Comparison of ratios, Ratio, Integral, Raabe's, Logarithmic and Cauchy's nth root (all tests without proofs), Alternating series, Absolute convergence, Conditional convergence. Function of Single

UNIT-II

Variable: Hyperbolic functions, inverse hyperbolic function, successive differentiation, Leibniz theorem, Taylor's and Maclaurin's theorems (without remainder terms).

UNIT-III

Curvature: Polar Curves, Differential coefficients of length of arc, Cartesian, polar and parametric forms, pedal equation, Angle between tangent and radius vector, Curvature and Radius of Curvature in Cartesian, polar and pedal forms.

UNIT-IV

Applications of definite integral: Asymptotes (in Cartesian), elementary knowledge of curve tracing, area, length, surface area and volume of revolution (in Cartesian, parametric and polar co-ordinates).

UNIT-V

Matrices: Elementary row transformation, Rank of matrix, consistency and inconsistency of system of simultaneous equations, solution of non-homogeneous and homogeneous equations, Eigenvalue and Eigenvector, Characteristic equation, Cayley-Hamilton theorem. Modal matrix

SUGGESTED READINGS:

1. Calculus and Analytic Geometry by G.B. Thomas (Pearson Education)
2. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Publication)
3. Advanced Engineering Mathematics by Michael Greenberg (Pearson Education)
4. Advanced Engineering Mathematics by R. K. Jain and S.R.K. Iyenger (Narosa Publication)
5. Higher Engineering Mathematics by B. S. Grewal (Khanna Publication)

Course No.	Title of the Course	Course Structure	Pre-requisite
FCCS002	Computer Programming	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

1. To understand the basic terminology and program structures used in computer programming to solve real world problems.
2. To understand the need for continuing to learn new languages to solve complex problems in different domains.
3. To learn the process of representing problems and writing, compiling and debugging programs.
4. To develop programming skills in using different types of data, decision structures, loops functions, pointers, data files and dynamic memory allocation/de-allocation.
5. To be able to code using Procedural and Object-Oriented languages.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Basics of C: Basic features of C Language like Identifier, Keywords, Variable, data types, Operators and Expression, basic screen and keyboard I/O, Control Statements, iteration, nested loops, Enumerated data types, bitwise operators, C Preprocessor statements.

UNIT-II

Arrays and Pointers: One and multidimensional dimensional arrays, strings arrays, operations on strings, Array and Pointers, Pointer to Pointer, other aspect of pointers, User Defined Data Types: Structures, Unions.

UNIT-III

Functions: Concept of modular programming, Using functions, Scope of data, Recursive functions, Pointers and functions, Command line arguments.

Files: Types of files, working with files, usage of file management functions.

UNIT-IV

Overview of Object Oriented Programming: Python Programming, Concepts and Terminology. Data Types and Collection Data Types: Identifiers and keyword, Integral types floating point types, operations and formatting, Sequence types, Tuples, named Tuples, lists, set Types, sets, frozen sets, mapping types, Dictionaries, Iterating and Copying collections iterators and interactable operations and functions copying collection.

Central Structures and Functions: Conditional branching, looping, Exception handling catching and raising exceptions, custom exceptions custom functions, Names and Docstrings, Argument and Parameter unpacking, Accessing variables in Global scope, lambda functions.

UNIT-V

Modules and Packages: Packages, custom modules, overview of python's standard library, string handling, mathematics and Numbers, Times and dates, File formats, Data persistence.

File Handling: Writing and Reading binary data, raw binary data, compression, parsing text files, Random Access binary files, generic binary record file class.

Guidelines for practical work:

Programs based on concepts of above languages.

SUGGESTED READINGS:

1. B. W. Kernighan and D.M. Ritchie, "The C programming language", Prentice Hall.
2. Herbert Schildt and Tata McGraw Hill, "The Complete Reference".
3. O Reilly Learning Python
4. Programming in Python 3: A Complete Introduction to the Python Language Pearson by Mark Summerfield

Course Type	Title of the Course	Course Structure	Pre-Requisite
FCEC003	**Electronics and Electrical Engineering**	**L-T-P: 3-0-2**	**None**

Course Outcomes (COs):

1. To understand the basics of AC and DC circuits, transformers along with DC generator and motor
2. To analyze series-parallel RLC circuits and
3. To implement basic circuits using diodes, BJTs and op-amps as circuit elements
4. To get familiarized with OP-AMP and its applications
5. To develop circuits using basic electrical and electronic components

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	2	2	-	-	-	-	-	-	-
CO 2	3	2	2	2	2	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	-	-	-	-	-	-	-
CO 4	3	2	2	2	2	-	-	-	-	-	-	-
CO 5	3	2	2	2	2	-	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Electric Circuits: Basic Circuit Elements, Nodal and Loop Analysis,

Superposition, Thevenin's Theorem & Norton's Theorem and Maximum Power Transfer Theorem;

UNIT-II

Steady-state analysis of AC circuits: Sinusoidal and phasor representation of Voltage and current, single phase AC circuit, behavior of R, L and C

Combination of R, L and C in series and parallel, Resonance; Introduction to three-phase circuits, Star-Delta Transformation

UNIT-III

Transformers: Principle of operation and construction of single-phase transformer, Introduction to DC Motor.

Electronics Devices and Circuits: Junction Diode, Applications: rectifiers, clipping and clamping circuits, LEDs;

UNIT-IV

Bipolar-junction Transistor: Physical operation, operating point, load-line, Self-bias circuit, single-stage CE amplifier configuration

Ideal op-amp, inverting, non-inverting and unity gain amplifiers, integrator, differentiator, summer/subtractor.

UNIT-V

Digital circuits- Boolean Algebra, logic gates, K-Maps upto 4-variables, Combinational circuits: Adders and subtractors.

Flip-Flops: SR, JK, D, T and their characteristic tables. Introduction to Sensors, Introduction to Embedded Computers.

LIST OF EXPERIMENTS:

1. Verification of Maximum Power Transfer theorem
2. Verification of Thevenin's and Norton's theorems
3. Study of resonance in series RLC and parallel RLC circuits
4. Analysis of step-up and step-down transformer
5. Implement of series RC circuit as differentiator and integrator. Also perform their analysis as low pass and high pass filters
6. Implementation of clipping and clamping circuits
7. Implementation of half-wave and full wave rectifier circuits
8. Application of LEDs in electronic circuits
9. Implementation of CE amplifying configuration. Plot gain vs frequency graph
10. Implementation of Adders and subtractors.
11. Implementation of JK and Toggle flip-flops. Subsequently implement 3-bit asynchronous up-counter.
12. Measurement of power in single phase circuits using three voltmeter and three ammeter method.
13. Experiments with common sensors
14. Experiment with embedded computers

SUGGESTED READING:

1. M.E. Van Valkenburg, "Network Analysis" Pearson publishers, 3rd Edition
2. Boylestad and Nashelsky, "Electronic Devices and Circuit Theory" Pearson publishers, 10th Edition
3. Edward Hughes, "Electrical and Electronic technology", Pearson publishers, 10th Edition
4. Malvino and Leach, "Digital Principles and Applications", TMH publishers, 8th Edition

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCPH004	**Physics**	**L-T-P: 3-0-2**	**None**

COURSE OUTCOMES (COs):

1. Knowing important concepts and phenomena linked to relativity
2. The concept of waves and oscillations are useful for doing analytical and numerical calculations for measurements, observations and gravitational wave communications.
3. The course is helpful to the students in understanding various optical wave phenomena which are required for optical & electromagnetic wave communications and in optical devices.
4. To develop the basic understanding of laser for gaining advance knowledge in the field of optical communication and opto-electronics.
5. The Concepts of Optical Fiber for modern developments in physics which are helpful in designing and developing new devices used in optical communications, medicine, environment, Industries and related physics.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PO 12
CO 1	3	2	2	2	2	1	-	-	-	-	-	-
CO 2	3	2	2	2	2	1	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	3	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Relativity: Special Relativity, Lorentz Transformations, Velocity addition, Time dilation, Length Contraction, Variation of mass with velocity, Mass and energy, Relativistic momentum and relativistic energy, General theory of relativity, Einstein's theory of Gravitation, Gravitational waves, Gravity and Light.

UNIT-II

Oscillations and Waves: Damped and forced oscillations, Sharpness of resonance, Q-factor, Application in resonance, Acoustic waves, Pressure wave equations, Intensity pressure relation, Acoustic impedance, Reflection and transmission of acoustic waves, Impedance matching; Ultrasonics and its applications.

UNIT-III

Optics: Interference: Interference due to thin films, Newton's rings, and determination of the wavelength of sodium light, Interference due to wedge shaped film. Diffraction: Fraunhofer diffraction due to single slit and N Slits, Plane transmission grating, Rayleigh criterion of resolution, Resolving power of a grating, Polarization: Polarization in light, Birefringence, Nicol

prism, Quarter and half wave plates, Production and analysis of plane, Circularly and elliptically polarized light, Optical rotation, specific rotation, Polarimeter.

UNIT-IV

Lasers: Absorption and emission of radiation, Main features of a laser, Spatial and temporal coherence, Einstein Coefficients, condition for light amplification, Basic requirement for Laser, Population Inversion - Threshold Condition, Line shape function, Optical Resonators, Three level and four level systems. Classification of Lasers: Solid State Laser-Ruby laser and Gas Laser- He-Ne laser (Principle, Construction and working), Optical properties of semiconductor, Semiconductor laser (Principle, Construction and working), Applications of lasers in the field of medicine, Industry, Environment and Communication.

UNIT-V

Fibre Optics: Need for fiber Optic Communication, Physical nature of Optical fiber, Theory of Light propagation in optical fiber, Acceptance angle and numerical aperture, Step index and graded index fibers, Single mode and multimode fibers, Losses in optical fiber, Optical Fiber cables and bundles, Dispersion in optical fibers: Intermodal and Intramodal dispersion.

LIST OF EXPERIMENTS:

1. To determine the value of "g" by Bar Pendulum and find the radius of gyration.
2. To determine the wavelength of He-Ne laser.
3. To find the numerical aperture and angle of acceptance of optical fiber.
4. To find the resolving power of a telescope.
5. To find the wavelength of sodium light by Newton's ring.
6. To find the wavelength of sodium light by Biprism.
7. To find the wavelength of Mercury green line by diffraction grating using spectrometer.
8. To find the focal length of combination of two lenses by Nodal slide assembly and verify formula.
9. To find the specific rotation of canesugar by polarimrter.
10. To find the dispersive power of prism material using spectrometer.

SUGGESTED READING:

1. Arthur Beiser, Shobhit Mahajan, `` Concepts of Modern Physics," Mc-Graw Hill
2. D S Mathur, ``Mechanics," S Chand & co.
3. N. Subramaniam and Brij Lal, ``A Text Book of Optics," S Chand &Co.
4. A K Jha "A Text Book of Applied Physics, Volume-1" I.K. International Publishing House.
5. Indu Prakash, ``A Text Book of Practical Physics, Volume-1," Kitab Mahal Publication.
6. Serwey, Moses, Moyer, ``Modern Physics," Cengage Learning
7. Jenkins and White, ``Fundamentals of Optics," McGraw Hill
8. Ajay Ghatak "Optics" McGraw Hill

Course No.	Title of the Course	Course Structure	Prerequisite
FCHS005	Core English	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

CO 1: Acquire competence in Basic English grammar. Grammatical accuracy, avoiding inappropriacy and using language naturally and confidently

CO 2: Improve in the four integral skills of language and to be able to use language as a tool for effective communication

CO 3: Enable the learner to express and be understood by others with clarity and precision, in both written and spoken forms, minimizing ambiguity and verbosity.

CO 4: Understand creative use of language through translation, articles and paragraph writing.

CO 5: Reading: Encouraging the habit of reading for different purposes and to analyse, paraphrase and read critically.

CO 6: Develop competence in formal Standard English pronunciation and usage

CO 7: Build confidence to use a standard spoken form of English to face job interviews, and workplace interactive situations besides enabling the learner to pursue advanced professional courses.

COs	Theory	Hours	Lab
1.	1. Vocabulary Enhancement CO 1 1.1 Using a standard dictionary- word spellings, meanings, usage, pronunciation, making sentences 1.2 Word collocations 1.3 Commonly misused words, verbal reasoning 1.4 One-word substitutions 1.5 Abbreviations & foreign phrases	4	Lab Activity No 1: Phonetics: Sounds Used in English Language CO 6 Lab Activity No 2: Reading from newspapers/magazines/blogs to build up a repertoire of words CO 5
2.	2. Remedial & Applied Grammar CO1 & CO 2 2.1 Tenses & Voice 2.2 Subject-Verb Agreement 2.3Narration, Interrogative structures and Question tags 2.4 Prepositions, Pronouns and Adverbs 2.5 Redundancy 2.6 Idiomatic use of language 2.7 Identification of errors and editing	6	Lab Activity No 3: Introducing Oneself: Breaking the Ice CO 5 Lab Activity No 4: Situational & Spontaneous English (tense, registers) through Role Play CO 7 Lab Activity No 5: Question Formation & Mock Press Conference CO 5

3.	3 Techniques of Good Writing CO 5 & CO 2 & 3 3.1Writing self-assessment tasks 3.2 Precis writing and note-making. 3.3 Paragraph and Essay writing. 3.4 Article writing and summarizing	10	Lab Activity No 6: Blog Writing/Creating a Newsletter Lab Activity No 7: Script writing & enacting for a street play CO 6
4.	4. Business Communication: CO 4 & CO 3 4.1 Formal and Informal Letter writing 4.2 Statement of Purpose 4.3 Job application & CV (summary statement of academic & professional profiles) 4.4Power point presentations through relevant slides.	10	Lab Activity No 7: Communication at Workplace. Develop negotiating skills by using appropriate language of courtesy Lab Activity No 8: Recording individual efforts and holding paired interactions and Group Discussions Lab Activity No 9: Preparing and practising for Interviews. CO 7
5.	5.Written Comprehension CO 3 & 4 5.1 The ability to write after listening to and reading select speeches, news bulletins, presentations and answering questions based on what has been heard. 5.2 Reading the given texts to skim, scan, infer and answer comprehension questions. 5.3 Reading texts like case studies and project reports for critical assessment. 5.4 Book Review	10	Lab Activity No 10:)Introduction to Podcast and Understanding Audio texts Lab Activity No 11: Declamation and/or speeches Lab Activity No 12: Reading, GD and presentation based on listed texts CO 7

Course No	Title of the Course	Course Structure	Pre-Requisite
FCME006	Basics of Mechanical Engineering	L-T-P: 4-0-0	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To know force, its nature and applications.
2. To know the basic principles of civil and mechanical structures.
3. To understand the fundamentals of thermodynamics and fluid mechanics.
4. To know the working principles of IC Engines.
5. To understand the importance of different engineering materials.
6. To understand the different manufacturing processes and machining operations.
7. To know the use of Automation in manufacturing.

PO / CO	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO11	PO12
CO 1	3	2	2	1	1	-	-	-	-	-	-	-
CO 2	3	2	2	1	1	-	-	-	-	-	-	-
CO 3	3	2	2	2	2	1	-	-	-	-	-	-
CO 4	3	2	2	2	2	1	-	-	-	-	-	-
CO 5	3	2	2	2	2	1	-	-	-	-	-	-
CO 6	3	2	2	2	2	1	-	-	-	-	-	-
CO 7	3	2	2	2	2	1	-	-	-	-	-	-

COURSE CONTENT:

UNIT-I

Introduction to Engineering Mechanics: Rigid and Elastic bodies, Force and its type, Law of parallelogram of forces, Triangle law of forces, Polygon law of forces, Lami's theorem, Laws of motion, Moment, Couple, Varignon's theorem, Conditions of equilibrium, Concept of free body diagram, Coulomb's friction, Plane trusses, Analysis of trusses, Numerical problems.

UNIT-II

Introduction to Strength of Materials: Simple stresses and strains, Direct, shear, and volumetric stresses and strains, Hooke's law, Tension test, Elastic constants, Poisson's ratio, Factor of safety, Introduction to beam, Types of beams, Types of loads, Shear force and bending moment diagrams (SFD and BMD) for Simple and Cantilever beams under various loading conditions, Numerical problems.

UNIT-III

Introduction to Manufacturing Engineering: Classification and use of engineering materials, Basic principles and applications of methods of manufacturing such as casting, forming and joining; Working principles and applications of machining operations such as Turning, Thread cutting, Milling, Shaping, Grinding, etc., Use of automation in manufacturing.

UNIT-IV

Introduction to Thermodynamics: Thermodynamic system, Cycle, Path, Thermodynamic properties, Extensive and intensive properties, Thermodynamic equilibrium, Reversible and irreversible processes, isochoric, Isothermal, Isobaric, Isentropic and Polytropic processes, First law of thermodynamics applied to a cycle and process, Kelvin-Planck and Clausius statements of Second law of thermodynamics, Carnot cycle, Entropy, Clausius inequality, Internal combustion (IC) engines, IC engines terminology, Spark ignition (SI) and Compression ignition (CI) engines, Two and four stroke engines, Air standard cycles such as Otto, Diesel, Dual and Brayton cycles, Numerical problems.

UNIT-V

Introduction to Fluid Mechanics: Properties of a fluid, Density, Specific volume, Specific weight, Specific gravity, Kinetic and Kinematic viscosity, Pascal's law and its applications, Laminar and turbulent flow, Use of continuity equation and Bernoulli's equation, Numerical problems.

SUGGESTED READINGS

1. Engineering Mechanics- Beer and Johnston, Pearson
2. Strength of Materials- D.K. Singh, CRC Press
3. Engineering Thermodynamics- Nag, McGraw-Hill
4. Fluid Mechanics- Cengel, McGraw-Hill
5. Fundamentals of Manufacturing Engineering- D.K. Singh, CRC Press

Course No.	Title of the Course	Course Structure	Pre-Requisite
FCMT007	Mathematics II	L-T-P: 3-1-0	None

COURSE OUTCOMES (COs):

1. Ordinary Differential Equations.
2. Partial Derivatives, Maxima and Minima for functions of two or more variables,
3. Evaluation of double and triple integral.
4. Concept of Numerical Methods and its Applications.
5. Concept of Probability and Statistics and its Applications.

COURSE CONTENT:

UNIT-I

Ordinary Differential Equations:

Second & higher order linear differential equation with constant coefficients, general solution of homogenous and non-homogenous equations, Euler-Cauchy equation, Series solution by Frobenius method.

UNIT-II

Function of Several Variables:

Partial Derivatives, Euler's Theorem, Total differentiations, Change of Variables, Jacobian and its basic properties, Taylor's theorem, Maxima and Minima for functions of two or more variables, Lagrange's method of undetermined multipliers.

UNIT-III

Multiple Integrals:

Evaluation of double integral (in Cartesian and polar co-ordinates), change of order of integration, change of variables. Triple integral (in Cartesian) and its applications. Gamma and beta function.

UNIT-IV

Numerical Methods:

Solution of system of linear equations using Gauss elimination method, LU decomposition method Gauss Seidel iteration method, Solution of polynomial and Transcendental equations by Newton-Raphson method, Numerical Integration by trapezoidal rule and Simpson's 1/3 and 3/8 rule, Numerical Solutions of first order ordinary differential equations: Euler's method, Runge-Kutta method of fourth order.

UNIT-V

Probability and Statistics:

Conditional probability, Random Variables, Probability distribution functions-binomial, Poisson, exponential, uniform and normal distributions; Correlation, rank correlation and regression analysis; Sampling Theorem.

SUGGESTED READINGS:

1. Calculus and Analytic Geometry by G.B. Thomas (Pearson Education)
2. Advanced Engineering Mathematics by Erwin Kreyszig (Wiley Publication)
3. Advanced Engineering Mathematics by Michael Greenberg (Pearson Education)

4. Advanced Engineering Mathematics by R. K. Jain and S.R.K. Iyenger (Narosa Publication)
5. Higher Engineering Mathematics by B. S. Grewal (Khanna Publication)
6. Probability and Statistics for Engineers by Anthony J. Hayter (Cengage Learning)
7. Numerical Methods for Scientific and Engg. Computations by M. K. Jain, S. R. K. Iyenger and R. K. Jain, (Wiley Eastern Ltd.)

4.2 FOUNDATION ELECTIVE COURSES

4.2.1 LIST OF FOUNDATION ELECTIVES

Table 3: List of Foundation Electives Courses

Code	Name of Foundation Elective	L T P Allocation			Evaluation Scheme					Pre-Requisites
					Theory			Practical		
		L	T	P	CA	MS	ES	CA	ES	
FEPD001	Sports-I	0	0	4	-	-	-	100	-	None
FEPD002	Sports-II	0	0	4	-	-	-	100	-	FE001
FEPD003	NSS	0	0	4	-	-	-	100	-	None
FEPD004	NCC	0	0	4	-	-	-	100	-	None
FEMG005	Corporate Social Responsibility	2	0	0	100	-	-	-	-	None
FEPD006	Music	0	0	4	-	-	-	100	-	None
FEHS007	Basic of social sciences	2	0	0	100	-	-	-	-	None
FEHS008	Spoken Skills in English	0	0	4	-	-	-	100	-	None
FEMG009	Financial Literacy	2	0	0	100	-	-	-	-	None
FEHS010	Introduction to Ethics	2	0	0	100	-	-	-	-	None
FEHS011	Stress Management	1	0	2	50	-	-	50	-	None
FEHS012	Organizational Behavior	2	0	0	100	-	-	-	-	None
FEPD013	Theatre	0	0	4	-	-	-	100	-	None
FEPD014	Dance	0	0	4	-	-	-	100	-	None
FEPD015	Yoga	0	0	4	-	-	-	100	-	None
FEPD016	Digital Film Making	0	0	4	-	-	-	100	-	None
FEPD017	Workshop (Electrical and Mechanical)	0	0	4	-	-	-	100	-	None
FEHS018	Ethical Decision Making	2	0	0	100	-	-	-	-	None

4.2.2 SYLLABI OF FOUNDATION ELECTIVES

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD001	Sports-I	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):
To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.

COURSE CONTENT: (Any 2 out of 5 Components)

A. INTRODUCTION TO PHYSICAL EDUCATION IN THE CONTEMPORARY CONTEXT (Any Two)

1. Learn and demonstrate the technique of Suryanamaskar.
2. Develop Physical Fitness through Calisthenics / Aerobics / Circuit-Training / Weight-Training and demonstrate the chosen activity.
3. Select any one game available in the college and learn different techniques involved in its play

B. CORE PHYSICAL EDUCATION-: FITNESS, WELLNESS AND NUTRITION (Any Two)

1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups Muscular Endurance); Harvard Step Test, Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measuring height, weight, waist circumference and hip circumference, Calculation of BMI (Body Mass Index) and Waist-Hip Ratio
3. Engage in at least one wellness programme and write a report on it.

C. CORE PHYSICAL EDUCATION-: POSTURE, ATHLETIC CARE AND FIRST AID (Any Two)

1. Demonstrate Stretching and Strengthening Exercises for Kyphosis, Scoliosis, Lordosis, Knock Knees, Bow Legs, Flat Foot, Back Pain and Neck Pain
2. Illustration and Demonstration of Active and Passive Exercises
3. Asanas with Therapeutic Value (Any five asanas): Karnapeedasana, Padmasana, Dhanurasana, Sarvangasana, Paschimottanasana, Chakrasana, Halasana, Matsyasana, Ardhmatsyendrasana, Usthrasana, Mayurasana, Shirshasana, Vajrasana.
4. Practice P.R.I.C.E. in First Aid.

D. SPORTS ADMINISTRATION & MANAGEMENT (Any Two)

1. Demonstration of Supervision activities in Sports Management.
2. Demonstration of skills of Management.
3. Demonstration of fixtures of various kinds in sports competitions.
4. Demonstration of technical and non-technical purchase procedure.

E. Adventure Sports and Leadership Training

SUGGESTED READINGS:

1. Graham, G., ``Teaching Children Physical Education: Becoming a Master Teacher. Human Kinetics," Champaign, Illinois, USA.
2. Corbin, C. B., G. J. Welk, W. R Corbin, K. A. Welk, ``Concepts of Physical Fitness: Active Lifestyle for Wellness," McGraw Hill, New York, USA.
3. Anspaugh, D.J., G. Ezell and K.N. Goodman, `` Teaching Today Health," Mosby Publishers
4. Beotra, Alka, ``Drug Education Handbook on Drug Abuse in Sports," Applied Nutrition Sciences, Mumbai.
5. Ammon,R., Southall , R.M. and Blair, D.A., ``Sports Facility Management, "West Virginia, USA: Fitness Information Technology Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD002	Sports-II	L-T-P: 0-0-4	FE001

COURSE OUTCOMES (COs):

To evolve a higher education system that is suitability blended with provision for knowledge values and skill practice where every student learns in without sacrificing his/her creativity.

COURSE CONTENT: (Any 3 out of 5 Components)

A. Sports for all (Any Two)

1. To participate in any intramural Tournaments (one team game and one Individual Game) of choice.
2. To participate/ attend at least 15 hours in Fitness training at Field or at Gymnasium.
3. Participate in at least one track and one field event on Annual Sports day.
4. To participate in Inter College Tournament

B. Skill Enhancement Courses (any one out of three)

1. Wellness and Fitness
2. Holistic personality Development
3. Sports Journalism
4. Mass demonstration Activities

C. MEDIA AND CAREERS IN PHYSICAL EDUCATION (Any Two)

1. Organize an event / intramural / tournament in your college.
2. Prepare a News Report of an observed Sports competition.
3. Create a presentation on any topic from Physical Education using an audio-visual aid.
4. Demonstrate Warming-up / Conditioning / Cooling-down exercises.

D. MANAGEMENT OF AEROBICS & GROUP TRAINING (Any Two)

1. Measurement of Fitness Components – Leg-raise for Minimal Strength (Muscular Strength); Sit-ups (Muscular Endurance); Harvard Step Test or Run and Walk Test (Cardiovascular Endurance); Sit and Reach Test (Flexibility)
2. Measurement of Pulse Rate / Heart Rate at Radial Artery and Carotid Artery, Calculation of Target Heart Rate
3. Developing a 5-10 minute routine of aerobics with appropriate music for each component of health related physical fitness

E. SPORTS INDUSTRY & MARKETING (Any Two)

1. Identify an issue or a trend in the sports industry: o Players in professional or college sports o Ownership
2. Marketing Plan: Environmental Factors and Product Plan Draft, Paper bibliography/works cited.
3. Sponsorship proposal
4. Developing a budget plan for an event
5. Athlete branding

SUGGESTED READINGS:

1. Covey, S. , `` 7 Habits of Highly Effective People, '' Covey Publications, USA
2. Magill, R.A., `` Motor Learning and Control: Concepts and Applications," McGraw Hill Publication.
3. Masteralexis, L.P., C. Barr and M. Humms, ``Principles and Practices of Sport Management," Jones and Bartlett Publisher

4. Bishop, J.G., ``Fitness through Aerobics,'' Benjamin Cummings USA.
5. Brown K.M., `` Physical Activity and Health: An Interactive Approach,'' Jones and Bartlett Publisher
6. Cornwell. T.B, `` Sponsorship in marketing: Effective communications through sports, arts and events, '' Routledge Publishers
7. DeGarris, L., ``Sports Marketing: A Practical Approach,'' Routledge Publishers, USA

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD003	**National Service Scheme (NSS)**	**L-T-P: 0-0-4**	**None**

COURSE OUTCOMES (COs):

1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;
4. Utilize their knowledge in finding practical solution to individual and community problems;
5. **Develop capacity to meet emergencies and natural disasters**

COURSE CONTENT:

Unit-I

Introduction to NSS: Orientation and structure of NSS, History of Social Reforms in Modern India: Brahmo Samaj, Arya Samaj, Satya Shodhak Samaj: Principles and Functions

Unit-II

Regular Activities: Distribution of working hours- association between issues and programs- community project- urban rural activities, association- modes of activity evaluation

Unit-III

Concept of Society- development of Indian society: Features- Division of labors and cast system in India; Features of Indian constitution; Provisions related to social integrity and development

Unit-IV

N.S.S. Regular Activities
A) College campus activities
B) N.S.S.activities in Urban and Rural areas
C) Role of Non-Government Organisation (NGO) in social Reforms
i) Red Cross
ii) Rotary

SUGGESTED READINGS:

1. National Service Scheme Manual, Govt. of India
2. Training Programme on National Programme scheme, TISS.
3. Orientation Courses for N.S.S. programme officers, TISS.
4. Ram Ahuja, ``Social Problems in India,'' Rawat Publication.
5. History of Social Reforms in Maharashtra, Ed. J. Y. Bhosale, S. U. Kolhapur

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD004	National Cadet Corps (NCC)	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

1. Develop among them a sense of social and civic responsibility;
2. Utilize their knowledge in finding practical solution to individual and community problems;
3. Identify the needs and problems of the community and involve them in problem solving process;
4. Utilize their knowledge in finding practical solution to individual and community problems;
5. Develop capacity to meet emergencies and natural disasters;

COURSE CONTENT:

UNIT-I

Introduction to NCC: National Integration & Awareness: Religions, Culture, Traditions and Customs of India, National Integration: Importance and Necessity, Freedom Struggle.

UNIT-II

Adventure Training: Obstacle course, Slithering, Trekking, Cycling, Rock Climbing, Para Sailing, gliding, Scuba Diving- methods and use.

UNIT-III

Environment Awareness and Conservation: Natural Resources - Conservation and Management. Water Conservation and Rainwater Harvesting

UNIT-IV

Personality Development and Leadership: Introduction to Personality Development, Factors Influencing /Shaping Personality: Physical, Social, Physiological, Philosophical and Psychological, Self-Awareness Know yourself/ Insight, Change Your Mind Set, Communication Skills: Group Discussion / Lecturettes (Public Speaking), Leadership Traits, Types of Leadership

SUGGESTED READINGS:

1. Bhogle Anita & Bhogle Harsha, ``The Winning way, Learning from sports for managers,'' Westland Publications
2. Sharma Robin, `` The leader had no title, '' Simon and Schuster Ltd.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG005	Corporate Social Responsibilities	L-T-P: 2-0-0	None

COURSE OUTCOMES (COs):

1. The course will help students to understand corporate and emerging social responsibility for the corporate in reference to India and global situation
2. The course will support students to prepare themselves to work with corporate understanding collective aspiration of the society, individual and corporate social responsibility.

COURSE CONTENT:

UNIT-I

Corporate social responsibility in Indian context and International: CSR – Definition, concepts, Approaches of CSR, overview of corporate social responsibility and corporate social accountability, SR Tools, National and International CSR activities, corporate philanthropy, drivers of CSR, difference between corporate governance, corporate philanthropy and CSR

UNIT-II

Business ethics and corporate social responsibility: Concept of business ethics – meaning, Importance and factors influencing business ethics. Corporate Governance – meaning, significance, principles and dimensions. Ethical decision – making in different culture, consumer protection, environment protection, gender issues in multiculturalism, ethics and corruption, ethics and safety. Business benefits of CSR

UNIT-III

Legislative measures of CSR: Corporate, labor, stake holders, Environmental and pollution. Social Accounting, Social Auditing, SA: 8000 and Corporate Social Reporting.

SUGGESTED READINGS:

1. Harsh Srivastava,`` The business of social responsibility," books for change
2. CV. Baxi and Ajit Prasad,`` Corporate social responsibility – concepts and cases," Excel Books
3. Dr. M. Mahmoudi,`` Global strategic management," Deep & Deep Publications Pvt. Ltd.
4. S K. Bhatia,`` International Human resource management – Global perspective," Deep & Deep Publications Pvt. Ltd.
5. J.P. Sharma, ``Governance, Ethics and Social responsibility of business, " Ane books Ltd.
6. Kotler Philip and Lee Nancy, `` Corporate social responsibility; doing the most good for your company," John Wiley
7. Simpson, Justine and Taylor, John R, `` Corporate Governance Ethics and and CSR," Kogan Page Publishers

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD006	Music	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

The student will be familiarized with the basic terms used in Indian classical music. Also it familiarizes with the life history of some dignitaries in the field of music. This course also throws some light on the ancient music and its origins in India.

COURSE CONTENT:

Unit-1

Study of the following terms: - Mela (Thāt), ĀshrayRāga, Rāga, Lakshana, Shruti, Alankar, Gamak, Vadi-SamvādiAnuvādi-Vivādi, VakraSwara, Varjit-Swara.

Unit-2

Biographies & contributions of the following: - Jaidev, MansinghTomar, Abdul Karim Khan, Tyagaraja, Pt. Bhatkhande, Pt. Ravi Shankar

Unit-3

Study of following Rāgas&TālaRāga- Yaman, Jaunpuri, Khamaj. Tāla- Ektāl, Jhaptāl

Unit-4

Genaral discussion and definition of the following: -

a. Khyāl, MaseetKhani – Razakhani gat, Dhrupad, Tarana, Meend, Soot, Murki, Kan, Khatka, Krintan, Harmony, Melody.

b. Writing of Bhatkhande Swarlipi Paddhati.

c. Writing of Tālasand Compositions in Notation.

d. Detailed study of Rāgas (Rāga- Bihag, Malkauns, Vrindavani Sarang) and comparative study of Rāgas.

e. Essay, Shastriya Sangeet (Classical Music) & SugamSangeet(Light Music)

Unit 5

Vedic Music – Samvedic Sangeet, Swara, Vadya, Bhakti, Vikār. General study of Natyashastra, SangeetRatnakar.

SUGGESTED READINGS:

1. Vasant and Laxmi Narayan Garg,`` Sangeet Visharad," Sangeet Karyalay
2. Sarat Chandra Pranjpayee and Chowbhamda ,`` BhartiyaSangeetkaItihas," Surbharti Prakashan
3. Bharat Muni,`` NatyaShastra,"
4. Sharangdeva ,`` SangeetRatnakar,"
5. Sharad Chandra Pranjpayee ,`` Sangeet Bodh,"
6. Thakur Jaidev Singh ,`` Indian Music," Sangeet research academy
7. V. N. Bhatkhande,`` Mallika Part II & III," KramikPustak.
8. V. N. Patwardhan,`` RaagVigyan,"
9. RaginiTrivedi,`` Ragvibodha Mishrabani, Vol. I & II,"

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS007	Basics of Social Sciences	L-T-P: 2-0-0	None

COURSE OUTCOMES (COs):

Social science is a major category of academic disciplines, concerned with society and the relationships among individuals within a society. It in turn has many branches, each of which is considered a "social science".

COURSE CONTENT:

Unit-I

Economics, political science, human geography, demography and sociology.

Unit-II

Humanities, anthropology, archaeology, jurisprudence, psychology, history, and linguistic.

Unit-III

Political science, economics, sociology, international politics and scientific methodology.

SUGGESTED READINGS:

1. A.C. Kapoor, “Principles of Political Science,” S. Chand Publications
2. A.K. Sharma, “Issues in Social Demography,” Mittal Publications
3. Kathy S. Stolley, “The Basics of Sociology,” Greenwood Press.
4. Paul M. Muchinsky ,“Psychology Applied to Work,” Thomson Learning Inc

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS008	**Spoken Skills in English**	**L-T-P: 0-0-4**	**None**

COURSE OUTCOMES (COs):

1. This course will focus on oral & presentation skills of students with practice sessions in the language lab.
2. This course will develop confidence building in oral skills of learners.
3. It will seek to encourage the day to day conversations/dialogues and communicative needs of learners with ample practice in the lab.
4. The theory class will boost practice in ample language exercises to encourage oral skills.
5. This will also involve practice sessions in interview skills, group discussions & pair work.
6. Basics of communication process, Barriers to Oral Communication
7. Elevator pitches - Practicals

COURSE CONTENT:

- Practice on listening and reading comprehension
- Language lab practice for group discussion and interviews
- Definition and discussion on communication & the barriers in communication with practical training to use language as a tool for sharing, discussing, handling and convincing others.

SUGGESTED READINGS:

Everyday English I & II Cambridge University Press/Foundation books

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEMG009	Financial Literacy	L-T-P: 2-0-0	None

COURSE OUTCOMES (COs):

1. To provide in-depth knowledge of the banking and Principles of Investment, financial planning.
2. Help students in understanding stocks, sell strategy, mutual fund options, investing in education, planning for the future, purchasing your first home, taxes and tax planning, life insurance options, health insurance, property insurance, estate planning, and keeping money in perspective.

COURSE CONTENT:

UNIT-I

Banking: Definition, Role of Bank in growth of saving and Investment, Types of banks , Services offered by banks, Deposits and Loans, Types of A/c, Opening a bank A/c, How to Transact with banks, KYC norms, (A/c opening form, Address Proof), How to read bank statement, Banking products and services, Calculating Interests – Saving, FD, Simple and Compound Interest, Power of compounding Loans, Types of loans, taking a home loan, Definition of EMI, Calculation of EMI, Post office-Account and transactions, Basic of foreign Exchange, Importance and Use of Foreign Exchange, Regulator Role of RBI, mutual funds.

UNIT-II

Investment: Principles of Investment – Safety, Liquidity and Return, Investment plans, Hybrid plans-Ulip, SIP and VIP of mutual funds, index funds

UNIT-III

Financial Planning: Meaning, Household financial health checkup, Important life stages, Medical and other Emergencies, ; Insurance, Meaning, Need and Wants, Loss protection, Life, non-life and health, Benefits of Insurance, Term plans, Social obligations Budgeting, Buying a house, Plan a vacation, Retirement planning, Price of procrastination, Market and financial instruments, Primary market, Secondary market, Financial Statement analysis,

UNIT-IV

Scams, Fraud Schemes: Insider trading, Money laundering; Consumer protection and redressal mechanism, Rights of Consumers, Applicable to financial services, Filing a complaint, Complain to entity concerned, Regulators, Arbitration, Consumer courts, Govt. Websites-(PG Portals), Investor Associations, Taxes, Meaning, Need of Taxes, Types of taxes, How taxes impact income, Income, wealth and gift tax, Service tax, STT, Stamp Duty, Tax planning v/s tax evasion, Tax rates, Tax free bonds, Tax saving investment

SUGGESTED READINGS:

1. Braunstein, Sandra, and Carolyn Welch, `` Financial literacy: An overview of practice, research, and policy," Fed. Res. Bull.
2. Cole, Shawn A., and Gauri Kartini Shastry,`` Smart money: The effect of education, cognitive ability, and financial literacy on financial market participation," Harvard Business School, 2009.
3. Study material of NSE.
4. Gitman, joehnk and Billingsley, ``Personal financial planning," Cengage Learning
5. Madura Jeff,`` Personal finance student edition,'' Prentice Hall PTR.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS010	**Introduction to Ethics**	**L-T-P: 2-0-0**	**None**

COURSE OBJECTIVES (COs):

1. It is aimed to comprehend right from wrong, to act upon something tricky with a deliberative analysis.

Course Outcomes:

2. Helps in addressing issues with a moral reasoning and analysis.

COURSE CONTENT:

1. Fundamentals of Ethics
2. Issues in Moral Philosophy
3. Theories of Justice and their Applications
4. Ethical Decision Making
5. Applied Ethics
6. Media Ethics
7. Environmental Ethics
8. Technology & Ethics
9. Feminism

SUGGESTED READINGS:

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS011	**Stress Management**	**L-T-P: 1-0-2**	**None**

COURSE OBJECTIVES (COs):

The objective of this course to help the students to understand the nature of stress, sources of stress and to identify the symptoms of stress through first unit. Second Unit aims to teach the students to learn certain skills and the strategies required for effectively managing the stress and ability to cope up from the stressful situations.

COURSE OUTCOMES (COs):

1. To understand the nature, sources of stress and consequences of stress
2. To overcome from the constraints in managing stress
3. To develop the motivation to adopt different technology

COURSE CONTENT:

UNIT-I

Stress (GAS Model), Learning about sources of stress and its symptoms: Nature of stress- various sources of stress environmental, social (including social media), physiological and psychological; Symptoms of stress - emotional response, physiological & behavioral; relationship between stress and performance, relationship between stress and health

UNIT-II

Learning to manage stress effectively: Methods - yoga, meditation, Vipassana, relaxation techniques, clarifying problem, alternate actions, support (Problem focused) emotion focused constructive approach, Indian Case Studies

Practical: (50 marks)

Any two practical's based on Unit I and II

SUGGESTED READINGS:

1. DiMatteo, M.R. & Martin, L.R.(2002). Health psychology. New Delhi: Pearson. Neiten, W. & Lloyd, M.A (2007). Psychology applied to Modern life. Thomson Detmar Learning .
2. Taylor, S.E. (2006). Health psychology, 6th Edition. New Delhi: Tata McGraw Hill

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS012	Organizational Behavior	L-T-P: 2-0-0	None

COURSE OUTCOMES (COs):

1. The course aims at providing a comprehensive understanding of organization (structure and culture) and its functioning, at the levels of the individual, group and organization as a whole.
2. To acquaint the students with employee motivation and work attitudes and its relationship with performance and productivity.
3. To help students gain insights into the concept of organization change in the context of ever changing business environment and provide them with various tools of organizational development.
4. To provide students an overview of organizational dynamics in the light of power, politics and stress.

COURSE CONTENT:

Unit-1

Understanding Organizational Behavior: Defining organization and Organizational Behavior (OB), OB Model; the Organizational structure; Common Organizational Designs; New Design Options

Unit-2

Employee Attitudes and Motivation: Job Satisfaction; Organizational Commitment; Organizational Citizenship behavior; Positive Organizational Behavior (POB):
Theories of Work motivation: Content theory (Maslow, Herzberg), Process theory (Vroom's Expectancy Theory, Equity Theory), Contemporary theories (Goal Setting theory and Self-Regulation theory)

Unit-3

Dynamics of Organizational Behavior: Organizational culture; Power and Politics: Influence, sexual harassment, organizational politics, Causes & Consequences of stress at the workplace:

Unit-4

Organizational change and development: concept of organizational change, model of organizational change (one model), organizational development: concepts, models (one model), techniques of organizational development

Unit-5

Organizational behavior in startups

SUGGESTED READINGS:

1. Aamodt, M. G. (2016). *Industrial/Organizational psychology: An applied approach*. Boston: Cengage Learning.
2. Luthans, F. (2011). Organizational behaviour:An evidence based approach,12th Edition. McGraw Hill
3. Muchinsky, P.(2007). Psychology applied to work: An introduction to Industrial and Organizational Psychology. NC: Hypergraphic Press.
4. Pareek, U. &Khanna,S.(2012,Third edition). Understanding Organizational Behaviour. Oxford: Oxford University Press.
5. Pareek, U., & Gupta, R. K. (2010).*Organizational behaviour*. New Delhi: Tata McGraw Hill.
6. Robbins, S.P., Judge T.A. and Sanghi, S.(2009) Essentials of Organizational Behaviour,10th edition. Pearson Education , India.
7. Schultz, D & Schultz, S (2013).Psychology and Work Today. Pearson Education, India.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD013	Theatre	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

Our goal is to nurture artist-scholars who are well read in dramatic literature, who understand the social and historical contexts of that literature, who appreciate contemporary performance and dance, who think critically, who master discipline-specific skills, and who make compelling artistic choices on stage.

COURSE CONTENT:

Unit-1

Concept of Acting in Indian Classical theatre. Western styles of theatre acting.

Unit-2

Basics of the following: Acting in Grotowski's Poor Theatre, Folk Theatre of India

Unit-3

Acting for Camera –Knowledge of camera frames and movement within the confines of a frame, blocking, difference between theatre and Camera acting, Concentration.

Unit-4

Acting consistently for different takes, acting scenes out of order, Auditions, acting exercises. Art of Dubbing.

SUGGESTED READINGS:

1. Boleslavsky, Richard,`` Acting: the First Six Lessons," New York Theatre Arts.
2. Hagen, Uta,`` Respect for Acting," Macmillan Press.
3. Hodge, Alison,`` Twentieth Century Actor Training," London and New York.
4. Routledge ,Stanislavski, Konstantin,`` An Actor's Work: A Student's Diary," Trans. and ed. Jean
5. Jeremiah Comey ,`` The Art of Film Acting," Focal Press .
6. Philips B Zarrilli,`` Acting (Re) Considered," Routeledge .
7. Cathy Hassey,`` Acting for Film," Allworth Press 9. Singh. Y,`` Indian Sociology social conditioning and emerging concerns," Vistaar publication.

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD014	Dance	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

This course will provide the student with the fundamentals necessary for advanced dance skills. Further, this course will develop student appreciation of dance as an art form and lifetime activity. Designed to familiarize students with technique, the student will also study vocabulary, different forms of dance, issues in dance and the history pertaining to the world of dance. The student will develop kinesthetic awareness, movement memory, creative abilities and aesthetic appreciation of various dance forms. The enhancement and the development and maintenance of physical fitness, self-confidence, self-discipline and independence with the body by providing informal showings during class are the goals expected to be achieved. Each student should leave this class having been encouraged, esteemed, and take with them a new appreciation of dance.

COURSE CONTENT:

- Basic workout
- Introduction to Hip Hop and B-Boying with a simple choreography
- Exercise like: Rolling, jumping, moving shoulders. Footwork, Floor steps, Beat knowledge.
- Freestyle combination along with House dance style.
- Expressions class: Body expressions, Face expressions.
- Introduction of Contemporary Dance. Basic exercise of Contemporary Dance. Exercise for flexibility, Floor steps, Spinning and Balancing.
- Introduction to Jazz. Basic exercise and proper routine practice.

SUGGESTED READINGS:

1. Jonathan Burrows, ``A Choreographer's Handbook,'' Routledge
2. Jacqueline M. Smith-Autard, ``Dance Composition: A Practical Guide to Creative Success in Dance Making,'' Routledge

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD015	Yoga	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

Students will learn about the importance of yoga in their lives. They will be exposed various types of yoga, their health benefits.

COURSE CONTENT:

UNIT-I

Origin of Yoga & its brief development, Meaning of Yoga & its importance, Yoga as a Science of Art (Yoga Philosophy),Meaning of meditation and its types and principles.

UNIT- II

Classification of Yoga/Types of Yoga, Hatha Yoga , Raja Yoga, Laya Yoga, Bhakti Yoga, Gyan Yoga, Karma Yoga, Asthang Yoga.

UNIT –III

Principles of Yogic Practices, Meaning of Asana, its types and principles, Meaning of Pranayama, its types and principles, Meaning of Kriya its types and principles.

UNIT -IV

Yogic therapies and modern concept of Yoga, Naturopathy, Hydrotherapy, Electrotherapy, Messotherapy, Acupressure, acupuncture, Meaning and importance of prayer, Psychology of mantras, Different mudras during prayers

SUGGESTED READINGS:

1. William Broad,`` The Science of Yoga: The Risks and the Rewards," Simon and Schuster
2. Swami Vishnu Devananda,`` The Complete Illustrated Book of Yoga," Harmony

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD016	Digital Film Making	L-T-P: 0-0-4	None

COURSE OUTCOMES (COs):

Students will learn about various technicalities involved in digital film making. They will also expose to history of cinema, preproduction etc.

COURSE CONTENT:

Unit-I

History of Cinema, Research & Scrip: Early Cinema, Development of Classical Indian & Hollywood Cinema, History of Global Film including European Film (1930-present), Origin of Classical narrative cinema-Soundless film, Exploration of film and analysis of the three-part beginning, middle and end of story, Research(Finding and Collecting materials and facts related to your story. Where and How to find the materials related to your story. Things to consider before sketching down your story), Script (Scriptwriting Process and its various phases), Film Grammar for Scriptwriting.

Unit-II

Pre-Production Digital Video Cinematography: Introduction to Digital Video Cinematography Cinematography, Interactivity and emotions through Cinematography,
Building blocks, Compositions, Lenses and Cameras, Types of lenses: Zoom Lens, Prime Lens, Types of Cameras: HD Cameras, Basics of Film Camera, Difference between, Film Camera and Digital Camera, DSLR and HDSLR Cameras, Lighting, Psychology of light, Visual Environment, Directional Effect of Light, Lighting design process, Three-point lighting, High-Key lighting, Low Key lighting, Construction of a Shot, Color, Contrast, Deep Focus, Shallow Focus, Depth of Filed, Exposure, Racking focus, Frame Rate, Telephoto shot, Zoom shot.

Unit-III

Digital Video Editing: Effective Editing, Principles of Video Editing, Non-Linear Editing (NLE) Concept, The Three-Point Edit, Non-Linear Editing (NLE) Techniques, Working in the Timeline, Transitions, Key framing, Applying Filters, Ingesting.

Unit-IV

Advanced Editing Techniques: NLE Compositing, Color Correction & Color Grading, Working on Audio, Titling

SUGGESTED READINGS:

1. Mark Brindle and Chris Jones, `` The Digital Filmmaking Handbook," Quercus

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEPD017	**Workshop (Electrical and Mechanical)**	**L-T-P: 2-0-0**	**None**

COURSE OUTCOMES (COs):

1. Student will be able to make various joints in the given object with the available work material.
2. The students will be able to understand various wiring connections

COURSE CONTENT:

Mechanical Workshop Experiments

1. Blacksmith
2. Carpentry
3. Fitting
4. Foundry
5. Welding

Electrical workshop Experiments

1. Study & Performance Of Different Types Of Wire Joints
2. Study And Performance Of Staircase Wiring
3. Study And Performance Of Series And Parallel Connection Of Flourescent Tube Light
4. Study And Performance Of Godown Wiring
5. Series And Parallel Connection Of Bulbs And Power Sockets By Single **Switch And Multi Switches.**

SUGGESTED READINGS:

1. Hajra Choudhury, Hazra Choudhary and Nirjhar Roy, ``Elements of Workshop Technology, vol. I, '' Media promoters and Publishers Pvt. Ltd.
2. W A J Chapman, Workshop Technology,`` Part -1, 1st South Asian Edition,'' Viva Book Pvt Ltd.
3. P.N. Rao, ``Manufacturing Technology, Vol.1,'' Tata McGraw Hill
4. Kaushish J.P., `` Manufacturing Processes, '' Prentice Hall

Course No.	Title of the Course	Course Structure	Pre-Requisite
FEHS018	**Ethical Decision Making**	**L-T-P: 2-0-0**	**None**

COURSE OUTCOMES (COs):

COURSE CONTENT:

UNIT-I

Ethical Concepts and Ethical Approaches:

1. Values, Dilemma and Choices
2. Responsibility, Justice & Fairness
3. Respect for self and others

UNIT-II

Ethical Decision Process:

1. Ethical codes and tests
2. Steps to ethical decision-making
3. Case studies and Situational role plays

SUGGESTED READINGS:

1. Blanchard, K., & Peale, N.V. (1988) The Power of Ethical Management, New York: William Morrow and Co. pp. 20-24. http://www.blanchardbowleslibrary.com/books/powerofethicalmanagement.htm
2. Brown, M. (1996) The Quest for Moral Foundations: An Introduction to Ethics Georgetown University Press
3. Davis, M. (1999) Ethics and The University, New York: Routledge.
4. Heller, R. (1998) Making Decisions, New York: DK.
5. Josephson, M. S. (2002) Making Ethical Decisions, Josephson Institute of Ethics.
6. Kardasz, F. (2008) Ethics Training For Law Enforcement: Practices and Trends, VDM, Verlag Dr. M.ller.
7. Nosich, G. M. (2002) Learning to Think Things Through: A Guide to Critical Thinking, Prentice Hall.

4.3 PROGRAM CORE COURSES

4.3.1 LIST OF PROGRAM CORE COURSES

SEM.	Code	Name of Core Course	Course Allocation			Evaluation Scheme Theory Practical					Syllabus page Nos
						Theory			Practical		
			L	T	P	CA	MS	ES	CA	ES	
II	MEMEC01	Engineering Mechanics	3	0	2	15	15	40	15	15	54
	MEMEC02	Engineering Materials & Metallurgy	3	0	2	15	15	40	15	15	56
	MEMEC03	Engineering & Machine Drawing	2	0	4	-	-	-	40	60	58
III	MEMTC04	Numerical Methods & Computation	3	1	0	25	25	50	-	-	60
	MEMEC05	Manufacturing Processes-I	3	0	2	15	15	40	15	15	61
	MEMEC06	Strength of Materials	3	1	0	25	25	50	-	-	63
	MEMEC07	Thermal Engineering-I	3	0	2	15	15	40	15	15	65
	MEMEC08	Fluid Mechanics & Machines	3	0	2	15	15	40	15	15	67
IV	MEMEC09	Theory of Machines	3	0	2	15	15	40	15	15	69
	MEMEC10	Manufacturing Processes-II	3	0	2	15	15	40	15	15	71
	MEMEC11	Thermal Engineering-II	3	0	2	15	15	40	15	15	73
	MEMEC12	Industrial Engineering & Management	3	1	0	25	25	50	-	-	75
	MEICC13	Control Systems	3	0	2	15	15	40	15	15	77

4.3.2 SYLLABI OF PROGRAM CORE COURSES : II SEMESTER

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC01	Engineering Mechanics	L-T-P:3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand the basic laws of engineering mechanics.
2. To analyze mechanics of rigid bodies and solve problems based on static equilibrium.
3. To understand the behaviour of rigid bodies under the friction force.
4. To locate the centroid/centre of gravity of rigid bodies.
5. To understand the behaviour of a particle under dynamic condition.

COURSE CONTENT:

UNIT-1

Basics and Statics of Particles

Basic laws of mechanics: Parallelogram, Triangular and Polygon law of forces, Principle of transmissibility of forces, Lami's theorem, System of forces, Equivalent system of forces, Vector representation of forces, Equations of equilibrium, Equilibrium of a particle.

UNIT-2

Equilibrium of Rigid Bodies

Free body diagram, types of supports and their reactions, Moments and Couples, Varignon's theorem, Equilibrium of rigid bodies in two dimensions and in three dimensions, Friction and its applications, Analysis of determinate trusses by analytical methods.

UNIT-3

Properties of Surfaces and Solids

Centroid, Centre of mass and Centre of gravity, Location of centroid/centre of gravity of different shapes, Moments of inertia: Area and mass moment of inertia of different bodies, Parallel and Perpendicular axis theorems, Pappus-Guldinus theorem, Moment of inertia of composite bodies.

UNIT-4

Kinematics of Particles

Rectilinear motion, Curvilinear motion, Rectangular, normal, tangential, polar, cylindrical, spherical coordinates, Relative and Constrained motion, Surface curvilinear motion, Force, mass and acceleration, Work and energy, Impulse and momentum, Impact.

UNIT-5

Kinematics of Rigid Bodies

Translation, fixed axis rotation, general planner motion, Work and energy, Power, Potential energy, Impulse-momentum and associated conservation principles, Euler equations of motion and its application.

LIST OF EXPERIMENTS:

1. To verify the law of moment by using a bell crank lever.
2. To verify the reactions at the supports of a simply supported beam.
3. To study the equilibrium of a particle under the action of forces in a plane (force table).
4. To determine the co-efficient of static friction between two given material surfaces with the help of an inclined plane.
5. To find the mechanical advantage, velocity ratio and efficiency of a simple screw jack.
6. To determine the effort required to lift a load and efficiency of lifting by wheel and differential axle.
7. To find the mechanical advantage, velocity ratio and efficiency of worm & worm wheel (Single start).
8. To find the mechanical advantage, velocity ratio and efficiency of worm & worm wheel (Double start).
9. To verify the forces in the members of a jib crane.
10. To construct a simple roof truss and find the forces in each member of the truss.
11. To determine the force in the members of a loaded shear legs space frame experimentally, vectorially and graphically.

Note: Minimum eight to ten experiments are to be conducted from the above list. Relevant Computer Simulation and Artificial Intelligence (AI) applications should be used in the listed experiments.

SUGGESTED READINGS:

1. Vector Mechanics for Engineers: Statics and Dynamics - Beer and Johnston, McGraw Hill.
2. Engineering Mechanics: Statics and Dynamics - Hibbeler, Pearson.
3. Engineering Mechanics: Statics and Dynamics - Shames, Pearson.
4. Engineering Mechanics: Statics - Meriam, Wiley.
5. Engineering Mechanics: Dynamics - Meriam, Wiley.
6. Engineering Mechanics - Pytel, Cengage Learning.
7. Engineering Mechanics - Harbola, Cengage Learning.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC02	Engineering Materials & Metallurgy	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To know the significance of material science and its role in manufacturing.
2. To classify the different engineering materials (metals, plastics, composites).
3. To describe the phase diagram and heat treatment processes.
4. To identify properties of engineering materials by various testing methods.
5. To develop concept of diffusion, mechanical properties and high temperature material problems.
6. To select a material for a specific use based on consideration of cost, performance and application.

COURSE CONTENT:

UNIT-1

Introduction: Historical perspective, importance of materials, crystallography and imperfections: Concept of unit cell, space lattice, Bravais lattices, common crystal structures, atomic packing factor and density. Mechanism of crystallization, Miller indices, X-ray crystallography techniques, crystal imperfections, Defects & Dislocations in solids, grain size.

UNIT-2

Mechanical Properties and Testing: Stress-strain curve, toughness, hardness, fracture, fatigue and creep. Destructive testing in respect of strength, hardness, impact, fatigue, ductility and creep. Ductile to brittle transition. Non-destructive testing (NDT).

UNIT-3

Phase Diagram and Equilibrium Diagram: Types of phase diagrams, types of microstructures, phase rules, types of equilibrium diagrams: solid solution types, eutectic type and combination type, Iron-carbon equilibrium diagram. Diffusion of Solids, Fick's I and II law.

Micro-Structural Examination: Microscope principle and methods, preparation of samples, micro-structural examination and grain size determination, comparative study of microstructure of various metals and alloys. Wear of metals, corrosion of metals.

Failure Analysis: Importance, procedure, modes of fractures, types of loadings, effect of stress raisers, strength reducers and residual stresses on material failure.

UNIT-4

Heat Treatment: Heat treatment processes, their needs, effects on micro-structure and applications, heat treatment of ferrous and nonferrous materials, case hardening processes and their applications, cooling curves, Time Temperature Transformation (TTT) diagrams.

UNIT-5

Composites: Classification, micro-mechanics of fibre and particle reinforced composites, strength, stiffness and factors affecting, failure modes.

Other Materials: Brief description of other industrial materials, Smart materials and nano-materials and other materials of industrial applications.

LIST OF EXPERIMENTS:

Minimum eight experiments are to be conducted from the following list:

1. Tensile test on a universal testing machine.
2. Compressive on a universal testing machine.
3. Torsion test of a rod on a torsion testing machine.
4. Creep test on a creep testing machine.
5. Fatigue test on a fatigue testing machine.
6. Hardness testing of a given specimen on Vickers/Brinell/Rockwell hardness testing machines.
7. Comparative study of microstructures of different specimens of different materials.
8. Study of heat treatment processes such as annealing, normalizing, quenching, case-hardening and comparison of hardness before and after heat treatment.
9. Study of nondestructive testing methods such as ultrasonic flaw detector, magnetic flaw detector and eddy current testing.

SUGGESTED READINGS:

1. Material Science for Engineering Students - Fischer, Academic Press.
2. Elements of Material Science & Engineering - Callister, Wiley.
3. Technology of Engineering Materials- Philip and Bolton, Butterworth - Heinamann.
4. Material Science - Raghvan, PHI.
5. Elements of Material Science & Engineering - Van Vlack, Pearson.
6. Material Science and Engineering - Smith and Hashemi, McGraw Hill.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC03	Engineering & Machine Drawing	L-T-P: 2-0-4	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To perform the free hand sketching of basic geometrical constructions.
2. To construct the orthographic projection of lines, planes and solids.
3. To understand section of solids and isometric projections of simple solids.
4. To apply the various concepts of engineering drawing like dimensioning, conventions and standards related to engineering and machine drawings.
5. To read and interpret the assembly drawings with moderate complexity.
6. To explain the conventions and the methods of assembly drawings.
7. To construct simple drawings using AutoCAD**.**

COURSE CONTENT:

UNIT-1

Projections: Free Hand and Orthographic Projections of Straight Lines, Planes and Solids, Section of Solids. Isometric Projection of Solids.

UNIT-2

Threaded Fasteners: Screw Thread: Terminology, Forms and their uses, Specifications of Bolts and Nuts, Types of Bolts and Nuts, Use of Washers, Special Purpose Nuts, Use of Stud & Tap Bolt, Locking arrangements for Nuts and Foundation Bolts.

UNIT-3

Rivets and Riveted Joints: Riveting, Rivet heads, Testing of Rivets, Caulking and Fullering, Failure of Rivets/Riveted joints.

UNIT-4

Cottered and Other Joints: Cotter, Gib, Types of Cottered joints, Knuckle joint.

Keys and Shaft Couplings: Key and Keyway, Types of Keys, Types of Shaft Couplings.

Bearings: Classification of Bearings, Bearing Materials, Prevention of Rotation of Brasses and Brackets.

UNIT-5

Pipe Joints: Pipe Materials and Applications, Types of Pipe Joints.

LIST OF EXPERIMENTS:

1. Free hand drawing sheets based on projections.
2. Drawing of threaded fasteners using CAD software.
3. Drawing of riveted joints using CAD software.
4. Drawing of Cottered and other joints using CAD software.
5. Drawing of key and shaft couplings using CAD software.
6. Drawing of bearings using CAD software.

7. Drawing of pipe joints using CAD software.

SUGGESTED READINGS:

1. Engineering Drawing- Bhatt, Charotar.
2. A Text Book of Engineering Drawing - Dhawan, S. Chand.
3. Engineering Graphics with AutoCAD - Naveen Kumar and S. C. Sharma, Galgotia.
4. Machine Drawing - S C Sharma, Standard Publishers Distributers.
5. A Text Book of Machine Drawing - Gill, S.K. Kataria & Sons.
6. Machine Drawing includes AutoCAD - Singh, McGraw Hill.

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4.3.3 SYLLABI OF PROGRAM CORE COURSES : III SEMESTER

Course No.	Title of the Course	Course Structure	Pre-requisite
MEMTC04	**Numerical Methods & Computation**	**L-T-P: 3-1-0**	**None**

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To create computer program for an algorithm and to use MATLAB.
2. To apply numerical methods for solving a system of linear and nonlinear algebraic equations.
3. To evaluate the roots of a function and to interpolate and extrapolate the given data.
4. To apply numerical methods for solving ordinary differential equations.
5. To solve partial differential equation using different numerical techniques.

COURSE CONTENT:

UNIT-1

Concepts of Computation: Concept of Algorithm and Computer Programming, Floating Point Operations and Errors, Revision of C++ and MATLAB, Taylor and Fourier Series Expansion.

UNIT-2

Linear and Nonlinear Algebraic Equations: Nonlinear System of Algebraic Equations, Conjugate Gradient Method, Bi-Conjugate Gradient Method, Tridiagonal Algorithm, Eigen Values and Eigen Vectors.

UNIT-3

Roots and Curves: Root Finding, Bisection Method. False Position Newton's Method, Secant Method, Interpolation, Extrapolation, Splines, Regression and Curve Fitting.

UNIT-4

Ordinary Differential Equations: Boundary-Value Problems and Initial Value Problems, Convergence and Error Studies, Single and Multistep Methods for Differential Equations, Simultaneous Ordinary Differential Equations.

UNIT-5

Partial Differential Equations: Classification of Partial Differential Equations, Solution of Parabolic, Elliptic and Hyperbolic Partial Differential Equations, Solution of Linear Partial Differential Equations by Finite Differences.

SUGGESTED READINGS:

1. Numerical Methods for Engineers - Chapra and Canale, McGraw Hill.
2. Introduction to Numerical Analysis - Shastry, PHI.
3. Numerical Methods for Engineers - Gupta, New Age International.
4. Applied Numerical Methods for Digital Computation - James, Smith and Wolford, Harper-Collins College Division.
5. Numerical Analysis and Computation: Theory and Practice - Blum, Addison-Wesley.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC05	**Manufacturing Processes - I**	**L-T-P: 3-0-2**	**None**

COURSE OUTCOMES (COs):

After completing this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand the basic manufacturing processes, and know about the different types of ferrous and nonferrous materials.
2. To know about casting, different casting processes and foundry tools used for the manufacturing of different products.
3. To know about joining, different welding processes and their applications.
4. To learn about the powder metallurgy processes and applications.
5. To know about sheet metal work and bench-work and fitting and their applications.

COURSE CONTENT:

UNIT-1

Introduction: Introduction and Importance of Manufacturing Processes, Classification and Overview of Manufacturing Processes; Ferrous Materials: Composition, Properties, and Applications of Cast iron, Wrought iron, Steel, Plain carbon steel, Alloy steel, Stainless steel, Tool steel, High speed steel; Nonferrous Materials: Composition, Properties, Alloys and Applications of Copper, Aluminium, Zinc, Tin, Lead, Nickel and Magnesium.

UNIT-2

Casting: Introduction to Metal Casting, Casting Terminology, Types of Patterns, Pattern Materials and Pattern Allowances, Classifications of Moulds, Types and Composition of Moulding Sand and their Desirable Properties, Gating System, Moulding Tools, Cupola Furnace, Types of Casting Processes such as Shell Casting, Slush Casting, Continuous Casting, Die Casting, Centrifugal Casting, Investment Casting, Casting Defects and Remedies.

UNIT-3

Metal Joining: Introduction, Classification of Welding Processes, Gas Welding: Types of Flames and their Applications, Arc Welding and Resistance Welding Processes, Solid State Welding, Welding Consumables, Soldering and Brazing, Hybrid Welding Processes, Welding Defects and Remedies.

UNIT 4

Powder Metallurgy: Introduction to Powder Metallurgy, Powder Metallurgy Processes, Methods of Powder Production, Blending, Compaction, Sintering, Applications of Powder Metallurgy, Advantages and Disadvantages of Powder Metallurgy in comparison to Conventional Shaping Processes.

UNIT-5

Sheet Metal Work: Tools and Equipment used in Sheet Metal Work, Sheet Materials, Standard Specifications for Sheets, Sheet Metal Operations.
Bench-Work and Fitting: Filing, Sawing, Chipping, Scraping, Internal and External Thread Cutting with Die, Hand Tools, Marking and Marking Tools.

LIST OF EXPERIMENTS:

1. Introduction to mechanical workshop.
2. Introduction to fitting shop and exercises involving fitting work.
3. Introduction to welding shop and preparation of simple butt and lap welded joints.
4. Introduction to sheet-metal shop and fabrication of sheet metal job such as funnel, tool-box, tray, electric panel box, etc.
5. Introduction to foundry shop and preparation of a sand mould with the use of a pattern.
6. Introduction to sand testing lab and sand preparation.

SUGGESTED READINGS:

1. Manufacturing Engineering & Technology - Kalpakjian, Pearson.
2. Fundamentals of Manufacturing Engineering - D. K. Singh, CRC Press.
3. Manufacturing Technology: Foundry, Forming and Welding - Rao, McGraw Hill.
4. Principles of Modern Manufacturing - Groover, Wiley.
5. Manufacturing Processes and Systems - Ostwald, Wiley.
6. Processes and Materials of Manufacture - Lindberg, Pearson.
7. Materials and Processes in Manufacturing - DeGarmo, Wiley.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC06	Strength of Materials	L-T-P: 3-1-0	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To know about stress and strain and their types, and applying the concepts in finding stress and elongation in prismatic and non-prismatic bars.
2. To know about simple bending, shear force and bending moment diagrams and deflection of beams.
3. To know about the torsion of solid and hollow circular shaft and spring.
4. To know about the various theories of failures and buckling of columns.
5. To know the stresses and strains produced in thin pressure vessels.

COURSE CONTENT:

UNIT-1

Analysis of Stress and Strain: Concept of Stress and Strain; Direct, Shear and Volumetric Stresses and Strains; Stress-Strain Curves (Engineering and Actual) for Ductile and Brittle Materials under Tension, Stress and Strain in Prismatic and Non-prismatic Bars under Axial Loading, Poisson's Ratio, Hooke's Law, Elastic Constants and their Relationships, Factor of Safety, Thermal Stress in Simple and Compound Bars, Plane Stress Condition: Principal Stress, Maximum Shear Stress, Mohr's Circle of Plane Stress.

UNIT-2

Theory of Simple Bending: Pure bending, Flexure formula, Section modulus, Flexural rigidity.

Shear Forces and Bending Moments in Beams: Beam and its types, Statically Determinate and Indeterminate Beams, Types of Loads on Beam, Shear Force and Bending Moment Diagrams (SFD and BMD) for Simple, Cantilever and Overhanging Beams, Point of Contraflexure.

Deflection of Beams: Differential Equation of Elastic Curve, Double Integration Method, Macaulay's Method, Deflection under various loading conditions for Simple and Cantilever Beams.

UNIT-3

Torsion: Pure Torsion of Solid and Hollow Circular Shafts, Torsion Formula, Polar Modulus, Torsional Rigidity, Power Transmitted by a Shaft.

Spring: Classification of Springs, Load-Deflection Curve, Close-Coiled Helical Spring subjected to an Axial Load.

UNIT-4

Theory of Elastic Failure: Maximum Normal Stress Theory, Maximum Normal Strain Theory, Total Strain Energy Theory, Maximum Shear Stress Theory, Maximum Distortion Energy Theory.
Columns: Introduction to column and its types, Column terminology, End conditions, Euler's Theory, Rankine-Gordon formula and other empirical formula for column.

UNIT-5

Pressure Vessels: Classification of Pressure Vessels, Stresses and Strains in Thin Cylindrical and Spherical Shells, Wire Wound Thin Cylinders.

SUGGESTED READINGS:

1. Strength of Materials - D.K. Singh, CRC Press.
2. Engineering Mechanics of Solids - Popov, Pearson.
3. Mechanics of Materials - Hibbeler, Pearson.
4. Strength of Materials - Timoshenko, CBS Publishers & Distributors.
5. Mechanics of Materials - Beer and Johnston, McGraw Hill.
6. Mechanics of Materials - Ugural, Wiley.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC07	Thermal Engineering - I	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand the basic concepts of thermodynamics such as temperature, pressure, system, properties, process, state, cycles and equilibrium.
2. To apply the knowledge of thermodynamic laws & processes to analyze air standard cycles used in internal combustion engines and gas turbines.
3. To get the knowledge of the use of steam tables and steam properties for applications in steam turbines, engines and thermal power plants.
4. To analyze the thermal power plants from electricity generation, economics and theoretical feasibility point of view.
5. To get the knowledge of other power plants such as gas turbine and nuclear power plants.

COURSE CONTENT:

UNIT-1

Introduction to Thermodynamics: Thermodynamic Terminology: Open, Closed and Isolated System, Cycle, Path, Process, Extensive and Intensive Properties, Thermodynamic Equilibrium; Zeroth Law of Thermodynamics, Absolute Temperature Scale, First law of Thermodynamics applied to a Process and a Cycle, Reversible and Irreversible Processes, Isochoric, Isothermal, Isobaric, Isentropic and Polytropic Processes, Kelvin-Planck and Clausius Statements of Second Law of Thermodynamics, Carnot Cycle, Reverse Carnot Cycle, Heat Pump and Refrigerator, Clausius Inequality, Entropy.

UNIT-2

Air Standard Cycles and Application: Otto Cycle, Diesel Cycle, Dual Combustion Cycle, Brayton Cycle, Internal Combustion (IC) Engines, IC engines Terminology, Spark Ignition (SI) and Compression Ignition (CI) Engines, Two-Stroke and Four-Stroke Engines, Knocking and Detonation.

UNIT-3

Properties of a Pure Substance: Phase Equilibrium of a Pure Substance on T-S Diagram, Normal Boiling Point of a Pure Substance, Saturation States, Compressed Liquid, *P-V* and *P-T* Diagrams of a Pure Substance, Dry and Saturated Steam, Superheated Steam, Use of Steam Tables and Mollier Diagram, Different Processes of Vapour on *P-V* and *T-S* Diagrams, Steam Quality.

UNIT-4

Thermal Power Plants: Simple Layout, Indian Scenario, Major Subsystems: Boiler, Turbine, Evaporator, Super-heater, Re-heater, Economizer, Dust Collectors, Ash Disposal, Fan and Draft Systems, Deaeration System, Cooling Water System and Cooling Towers.

UNIT-5

Other Power Plants: Brayton Cycle, Reheating, Intercooling, Regeneration, Efficiency and Work Output, Gas Turbine Plants and Fields of Application, Nuclear Power Plants, Power Reactors, Advantages and Disadvantages of Nuclear Power Plants.

LIST OF EXPERIMENTS:

1. To demonstrate the workdone due to heat transfer.
2. To prove the first law of thermodynamics for a closed system.
3. To find out the energy balance in an open system.
4. To plot a graph between Brake Horse Power and Specific Fuel Combustion.
5. To plot a graph between Brake Horse Power and Total Fuel Combustion.
6. To plot a graph between Brake Horse Power and Mechanical Efficiency.
7. To plot a graph between Brake Horse Power and Volumetric Efficiency.
8. To analyze the exhaust gases from an engine.
9. To measure the steam quality.

SUGGESTED READINGS:

1. Thermodynamics: An Engineering Approach - Cengel, McGraw Hill.
2. Fundamentals of Thermodynamics - Borgnakke and Sonntag, Wiley.
3. Fundamentals of Engineering Thermodynamics - Moran, Shapiro, Boettner and Bailey, Wiley.
4. Engineering Thermodynamics - Nag, McGraw Hill.
5. Power Plant Engineering - Nag, McGraw Hill.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC08	Fluid Mechanics & Machines	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To apply the theory of fluid mechanics to determine the forces on planar and curved surfaces submerged in a static fluid.
2. To analyze the fluid motion with or without the consideration of external forces.
3. To apply the knowledge of boundary layer flow and turbulent flow for flow through pipes.
4. To implement the momentum principle for fluid machines.
5. To know about the power transmission by hydraulic and pneumatic systems.

COURSE CONTENT:

UNIT-1

Fluid Properties: Fluid statics, Manometry, Buoyancy, Forces on submerged bodies, Stability of floating bodies; Control-volume analysis of mass.

UNIT-2

Fluid Kinematics and Dynamics: Classification of fluid flows, Flow lines, Continuity equation, Stream function, Potential function, Rotational flow, Rotation and Vorticity, Flow nets, Momentum and energy, Fluid acceleration, Differential equations of continuity and momentum, Bernoulli's equation, Dimensional analysis, Viscous flow of incompressible fluids, Venturimeter, Orifice meter, Rotameter and Mouthpieces, Pitot tube.

UNIT-3

Flow Through Pipes: Boundary layer flow, Elementary turbulent flow, Flow through pipes, Head losses in pipes, Bends and Fittings, Reynold's experiments, Prandtl mixing length hypothesis, Velocity distribution in pipes, Concept of smooth and rough pipes, Pipe friction factor relations. Pipe network.

UNIT-4

Fluid Machines: Impulse and Reaction principles, Velocity diagrams, Pelton wheel, Francis and Kaplan turbines, Centrifugal pumps, Turbine pumps, Axial and Mixed flow pumps, Performance and Characteristic curves, Water hammer, Turbine and Pump installations, Hydraulic ram.

UNIT-5

Power Hydraulics: Positive pumps, gear, vane, screw, variable delivery pumps, valves, Flow control, Pressure control, Direction control, Solenoid operated valve, Hydraulic circuits, (meter-in, meter-out, bleed-off), Fluid coupling and Torque converter, Principles of pneumatic power, Comparison of Pneumatic and Hydraulic Systems.

LIST OF EXPERIMENTS:

1. To study the laminar flow in a pipe and plot the velocity distribution.
2. To carry out the Reynolds experiment and observe the flow regimes.
3. To prove the Bernoulli's equation.
4. To find the coefficient of discharge of an Orifice meter.
5. To find the coefficient of discharge of a Venturimeter.

6. To find the coefficient of discharge of Notches.
7. To plot the performance characteristics of a centrifugal pump.
8. To plot the performance characteristics of a Pelton wheel.
9. To plot the performance characteristics of a Francis turbine.
10. To plot the performance characteristics of a Kaplan pump.

SUGGESTED READINGS:

1. Fluid Mechanics - Streeter, McGraw Hill.
2. Mechanics of Fluids - Shames, McGraw Hill.
3. Fluid Mechanics - Cengel, McGraw Hill.
4. Fluid Mechanics - White, Wiley
5. Fluid Mechanics and Fluid Power Engineering - D.S. Kumar, S.K. Kataria & Sons.

4.3.4 SYLLABI OF PROGRAM CORE COURSES : IV SEMESTER

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC09	Theory of Machines	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

1. To understand the function of a mechanism in common machine tools.
2. To know the kinetics and kinematics involved in the study of relative velocity and acceleration among the links of a mechanism.
3. To understand the inertia forces involved in an internal combustion engine.
4. To study the various possible motions between cam-follower and gears assemblies.
5. To study the natural frequency of undamped free vibration under different conditions.

COURSE CONTENT:

UNIT- 1

Introduction: Kinematic Links, Kinematic Pairs, Kinematic Chain, Types of Joint in a Kinematic Chain, Inversions of Four Bar Chain. Types of Mechanism, Degree of Freedom of Mechanisms in a Plane.

UNIT- 2

Velocity and Acceleration Analysis of Planar Mechanisms: Instantaneous Centre Method, Polygon and Graphical Methods, Coriolis Component of Acceleration.

UNIT- 3

Dynamics of Planar Mechanisms with Special Reference to Slider-Crank Mechanism and Internal Combustion Engines: Approximate Analytical Method for Velocity and Acceleration of the Piston, Angular Velocity and Angular Acceleration of the Connecting Rod, Turning Moment on the Crankshaft. Turning Moment Diagram for a Four-Stroke Internal Combustion Engine.

UNIT- 4

Kinematics of Higher Pair Mechanisms:
Cam: Nomenclature, Follower Motions, Graphical and Analytical Methods of Cam Profiles, Pressure Angle.
Gears: Introduction, Law of Gearing, Synthesis of Tooth Profile, Undercutting and Interference; Simple, Compound and Epicyclic Gear Trains, Differential Gear Box.

UNIT- 5

Introduction to Natural Frequency of Free Vibrations: Types of Free Vibratory Motions Longitudinal, Transverse and Torsional Vibrations, Critical or Whirling Speed of a Shaft.

LIST OF EXPERIMENTS:

1. To study the various inversions of a four-bar chain and find out the nature of motion in various links.
2. To trace the involute gear teeth profile using a rack cutter.
3. To plot the Follower displacement vs. Cam angle rotation; Study of Velocity, Acceleration and Jerk w.r.t. Follower displacement.
4. To study the function of a governor and to obtain a graph between Force vs. Radius of rotation by using Watt and Hartnell governors.
5. To study the gyroscopic behaviour and verify gyroscopic couple due to load applied.

6. To study the undamped free vibration with respect to:
   (a) Pendulum apparatus
   (b) Radius of gyration of a given pendulum
   (c) By-filer suspension
   (d) Longitudinal vibrations of a helical spring.
7. To determine the torque distribution in an epicyclical gear train and to study the functioning of a hand operated model.
8. To study the "Whirling of shaft".
9. To study the "Balancing of Rotary Masses".
10. To study the working of Differential gear mechanism.

SUGGESTED READINGS:

1. Theory of Machines and Mechanism - Shigley and Joseph, Oxford University Press.
2. Theory of Machines and Mechanisms - Ballaney, Khanna Publishers.
3. Theory of Machines - Khurmi and Gupta, S. Chand & Company Ltd.
4. The Theory of Machines - Bevan, CBS Publication.
5. Theory of Machines - Ratan, McGraw Hill.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC10	Manufacturing Processes-II	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand the concepts of metal cutting and its important role in manufacturing processes.
2. To learn about the tool materials and their effects on tool performance and cutting operation.
3. To understand the working of different machine tools along with their right usage.
4. To understand the need and importance of finishing operations.
5. To learn about the specific product manufacturing like gears and importance of accurate positioning during machining and inspection.
6. To understand the principle and working of different non-traditional machining processes.
7. To utilize the knowledge gained in completing the academic and industrial assignments and projects successfully.

COURSE CONTENT:

UNIT-1

Metal Cutting: Mechanics of Metal Cutting, Single Point Cutting Tool Geometry and Nomenclature, Orthogonal and Oblique Cutting, Mechanics of Chip Formation, Types of Chips, Shear Angle Relationship, Merchant's Force Circle Diagram, Cutting Forces and Power required, Cutting Fluids/Lubricants, Tool Materials, Tool Wear and Tool Life, Measurement of Cutting Forces, Economics of Machining.

UNIT-2

Machine Tools: Construction, Types, Important Mechanisms, Working Principle, Operations, Accessories, Attachments and Applications of Lathe, Drilling Machine, Milling Machine, Shaper, Slotter, Planer, Broaching and Boring Machines, Multiple Point Cutting Tool Geometry and Types, Dividing Head, Types of Indexing, Installation of Machine Tools and Safety Measures.

UNIT-3

Grinding and Super Finishing: Introduction, Construction, Types, Working Principle, Operations, Accessories, Attachments and Applications of Surface, Cylindrical, Internal and Centerless Grinding Machines; Tool and Cutter Grinding, Grinding Wheel Specifications, Shapes, Setting, Dressing and Truing, Grinding Ratio, Super Finishing Operations: Honing, Lapping, and Polishing.

UNIT-4

Gear Cutting: Types of Gears and Terminology, Gear Generation Processes, Gear Finishing Processes, Gear Inspection, Introduction to Jigs and Fixtures.

UNIT-5

Non-Traditional Machining: Introduction, Need of Non-traditional Machining Processes and their Classification; Principle, Working and Applications of Chemical Machining, Electrochemical Machining (ECM), Abrasive Jet Machining (AJM), Ultra-Sonic Machining (USM), Electro Discharge Machining (EDM), Electron Beam Machining (EBM), Laser Beam Machining (LBM), Electrochemical Grinding (ECG).

LIST OF EXPERIMENTS:

1. Introduction to the machine shop and its facilities.
2. Demonstration of the lathe machine, its various functions and capabilities.
3. To center a given job on a four-jaw chuck.
4. To machine the given job on a center lathe involving turning, facing, step turning, grooving, taperturning, drilling, knurling and champhering.
5. To perform eccentric turning on a center lathe.
6. To perform plain milling, face milling, end milling, saw milling and key cutting on a millingmachine.
7. To perform indexing and spline cutting on a milling machine.
8. Introduction to shaper and to perform plane shaping operation.
9. To perform shape cutting on a shaper.
10. To perform slot cutting on a shaper.

SUGGESTED READINGS:

1. Manufacturing Engineering and Technology - Kalpakjian, Pearson.
2. Fundamentals of Manufacturing Engineering - D. K. Singh, CRC Press.
3. Principles of Modern Manufacturing - Groover, Wiley.
4. Manufacturing Technology, Vol-2, - Rao, McGraw Hill.
5. Materials and Processes in Manufacturing - De Garmo, Wiley.
6. Manufacturing Processes and Systems - Ostwald, Willey.
7. Manufacturing Science - Ghosh and Malik, Affiliated East West Press.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC11	Thermal Engineering - II	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To apply the theory of conduction to design devices, such as thermal insulation and extended surfaces.
2. To get the knowledge of heat convection, boundary layer concept, condensation and boiling.
3. To get the knowledge of the fundamentals of heat radiation, heat exchange between bodies and solar radiation.
4. To apply the heat transfer theories to design heat exchanger for various applications.
5. To get the knowledge of vapour compression cycle, air conditioners and heat load calculation.

COURSE CONTENT:

UNIT-I

Conduction: One Dimensional Steady State Conduction, Simple Convection. Overall Heat Transfer Coefficient, Simple Cases of Heat Transfer through Homogenous and Composite Plane Walls, Cylinders and Spheres with Constant and Variable Thermal Conductivity, Critical Thickness of Insulation, Heat Transfer from Fins of Uniform Cross Section.

UNIT-II

Convection: Introduction to Hydrodynamic and Thermal Boundary Layers, Application of Dimensional Analysis to Free and Forced Convection, Film Condensation and Dropwise Condensation, Heat Transfer Coefficient, Boiling Heat Transfer.

UNIT-III

Radiation: Kirchhoff's Law, Planck's Distribution Law, Wein's Displacement Law, Stefan-Boltzmann's Relation and Configuration Factor, Radiant Heat Interchange between Black and Grey Surfaces, Solar Radiation, Radiation Shielding.

UNIT-IV

Heat Exchangers: Parallel Flow, Counter Flow and Cross Flow Heat Exchangers, Shell and Tube, Finned Tube and Air Cooled Heat Exchangers; LMTD and NTU Methods, Exchangers Design, Simple Calculations.

UNIT-V

Refrigeration and Air-conditioning: Vapor Compression Cycle, Effect of Sub-cooling and Superheating, Components of a Typical Refrigerator and Air-conditioner: Compressor, Evaporator, Condenser and Expansion Valve, Vapor Absorption System, Domestic Refrigerator, Window and Split Air-conditioners, Heat Load Calculation.

LIST OF EXPERIMENTS:

1. To determine the thermal conductivity of the given composite walls.
2. To determine the Stephan-Boltzmann Constant experimentally.
3. To determine the heat transfer co-efficient by forced convection.
4. To determine the heat transfer co-efficient by natural convection.
5. To determine the overall heat transfer co-efficient of the shell and tube heat exchanger.
6. To measure the convective heat transfer co-efficient and effectiveness of the fin under forced andnatural convection.
7. To determine the heat transfer co-efficient for transient heat transfer apparatus.
8. To determine the critical radius of insulation for critical radius apparatus.
9. To determine the COP of a vapor absorption cycle.

SUGGESTED READINGS:

1. A Textbook on Heat Transfer - Sukhatme, University Press.
2. Heat and Mass Transfer: Fundamentals and Applications - Cengel, McGraw Hill.
3. Heat Transfer - Holman, McGraw Hill.
4. Heat and Mass Transfer - Nag, McGraw Hill.
5. Principles of Heat and Mass Transfer - Bergman, Lavine, DeWitt and Incropera, Wiley.
6. Refrigeration and Air Conditioning - Arora, McGraw Hill.

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEMEC12	Industrial Engineering & Management	L-T-P: 3-1-0	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand the importance of industrial engineering in industry.
2. To apply the industrial engineering concepts in the industrial environment.
3. To manage and implement different concepts involved in work study.
4. To identify various cost accounting and financial management practices widely applied in industries.
5. To understand the vital role of forecasting and its accuracy.
6. To know about the correct production planning to achieve organizational goals.
7. To know about the just in time production system.
8. To select and analyze an inventory control model.

COURSE CONTENT:

UNIT-1

Introduction: Definition, Objectives, Contributions, Activities and Techniques of Industrial Engineering; Industrial Engineering in Service Sector, Systemic Approach; Motivation and Theories of Motivation, Productivity and its Measurement, Factors Affecting Productivity and Methods to Improve Productivity; Plant Location and Layout: Objectives, Types, Advantages and Influencing Factors.

UNIT-2

Work Study: Introduction, Importance and Procedure of Work Study; Method (Motion) Study: Scope, Objectives and its Various Techniques, Critical Examination; Work Measurement (Time Study): Objectives and Techniques, Performance Rating, Allowances, Standard Time Computation, Work Sampling, Job Evaluation and Merit Rating, Wages and Incentive Schemes.

UNIT-3

Cost Analysis and Demand Forecasting: Fixed & Variable Costs, Direct, Indirect and Overhead Costs and Job Costing, Recovery of Overheads, Standard Costing, Cost Control, Break Even Analysis; Forecasting: Importance, Objectives, Forecasting and Prediction, Types, Classification of Forecasting Methods, Forecast Errors, Costs and Accuracy of Forecast.

UNIT-4

Production Planning & Control (PPC): Definition, Need, Objectives and Functions of PPC, Aggregate Planning, Materials Requirement Planning (MRP), Routing, Scheduling, Machine Loading using Johnson's Rule, Dispatching; Various Means of Measuring Effectiveness of PPC, Introduction to Just in Time Production System, Project Management: CPM and PERT.

UNIT-5

Materials Management: Definition, Strategic Importance of Materials in Manufacturing Industries, Relevant Costs, Basic Economic Order Quantity (EOQ) Model, Model with Quantity Discount, Production Quantity Model, Re-order Point and Order Quantity; Selective Inventory Control Techniques such as ABC, VED, FSN, XYZ, SDE and SOS Analysis.

SUGGESTED READINGS:

1. Operations Management: Processes and Supply Chains - Ritzman and Krajewski, Pearson.
2. Production and Operations Management - Render and Heizer, Pearson.
3. Operations Management - Stevenson, McGraw Hill.
4. Operations Management for Competitive Advantage - Jacobs and Chase, McGraw Hill.
5. Modern Production/Operations Management - Buffa and Sarin, Wiley.
6. Just in Time: Concept and Practices - D.K. Singh, Ane Books India.

NSUT
NETAJI SUBHAS UNIVERSITY OF TECHNOLOGY

Course No.	Title of the Course	Course Structure	Pre-Requisite
MEICC13	Control Systems	L-T-P: 3-0-2	None

COURSE OUTCOMES (COs):

After completion of this course, the students are expected to be able to demonstrate the following knowledge, skills and attitudes:

1. To understand and analyze the feedback characteristics of linear control system to reduce the disturbance.
2. To determine the response of different order systems for various standard inputs.
3. To perform the frequency domain analysis of linear control system using bode plot and Nyquist stability criterion.
4. To design various types of compensators.
5. To design state feedback controller.

COURSE CONTENT:

UNIT-1

Introduction: Open Loop and Closed Loop Control Systems, Block Representation, Linear and Non-linear Control, Continuous and Sampled Data Control, Digital Control.

Mathematical Models of Physical Systems: Mathematical Modelling and System Representation of Physical Systems, Analogous Systems. AC, DC servomotor, Tachogenerator.

Transfer Function, Block Diagram Representation and Reduction Technique, Signal Flow Graph-
Construction, Terminology, Algebra and Mason's Gain Formula.

Feedback Characteristics of Control Systems: Effects of Feedback on Variation of System Parameters, System Dynamics and Effect of Disturbances.

UNIT-2

Time Domain Analysis: Standard Test Signals, Transient Response for First and Second Order Systems, Transient Response Specifications, Effect of Proportional, Integral and Derivative Control Actions on System Performance, Performance Index Concept and Error Performance Indices- ISE, ITSE, IAE, ITAE, Concept of Stability, Conditions for Stability. Routh-Hurwitz Criterion, Root Locus Technique Concept, Construction Rules.

UNIT-3

Frequency Domain Analysis: Concept of Frequency Response, Frequency Response Plots, Polar Plot, Bode Plots, Performance Specifications, Correlation between Time and Frequency Responses.

Stability in Frequency Domain: Nyquist Criterion, Gain and Phase Margin.

UNIT-4

Compensation Techniques: Control Systems using Compensation Networks such as, Lag, Lead, Lag-lead Networks via Frequency Domain Techniques.

UNIT-5

Introduction to Modern Control: State Space Representations of Transfer Function Systems, State transition matrix, Controllability and Observability, Control with State Feedback.

SUGGESTED READINGS:

1. Modern Control Engineering - Ogata, Pearson.
2. Modern Control System - Dorf and Bishop, Pearson.
3. Automatic Control System - Kuo, Pearson.
4. Control Systems Engineering - Nagrath and Gopal, New Age International.

4.3.5 SYLLABI OF PROGRAM CORE COURSES : V SEMESTER

4.3.6 SYLLABI OF PROGRAM CORE COURSES : VI SEMESTER

4.4 DEPARTMENT ELECTIVE COURSES

4.4.1 LIST OF DEPARTMENT ELECTIVES

4.4.2 SYLLABI OF DEPARTMENT ELECTIVES COURSES : V SEMESTER

4.4.3 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VI SEMESTER

4.4.4 SYLLABI OF DEPARTMENT ELECTIVES COURSES : VII & VIII SEMESTERS

4.5 OPEN ELECTIVE COURSES

4.5.1 LIST OF OPEN ELECTIVES

4.5.2 SYLLABI OF OPEN ELECTIVE COURSES : SCIENCES

4.5.3 SYLLABI OF OPEN ELECTIVE COURSES : HUMANITIES & SOCIAL SCIENCES

4.5.4 SYLLABI OF OPEN ELECTIVE COURSES : MANAGEMENT